

```
// BootRunner.cs
using System.Collections;
using Sy.Core;
using UnityEngine;
using UnityEngine.SceneManagement;
using Logger = Sy.Core.Logger;
namespace Sy.P08.Boot
{
    public static class SceneIds
    {
        {
            public const string Boot = "S00_Boot";
            public const string Main = "S08_Main";
        }
        /// <summary>
        /// S00_Boot      Mock → →      T001④
        ///      Boot      UI T001b  Boot      =
        /// </summary>
        public sealed class BootRunner : MonoBehaviour
        {
            IEnumerator Start()
            {
                BootServices.RegisterAll();
                var sequence = new BootSequence(SwitchToMain);
                if (!sequence.RunAll())
                {
                    Logger.Error("BOOT", "boot sequence failed - stay on S00_Boot");
                    yield break;
                }
                yield return null;
            }
            bool SwitchToMain()
            {
                StartCoroutine(LoadMainRoutine());
                return true;
            }
            IEnumerator LoadMainRoutine()
            {
                {
                    var op = SceneManager.LoadSceneAsync(SceneIds.Main, LoadSceneMode.Single);
                    while (op != null && !op.isDone) yield return null;
                }
            }
        }
    }
}

// BootSequence.cs
using System;
using System.Collections.Generic;
using Sy.Core;
using Sy.Platform;
namespace Sy.P08.Boot
{
    /// <summary>
```

```

/// Boot      T001
/// ①Logger→② →③Config→④Save→⑤ILocale→⑥Ads.Preload→⑦      [BOOT] n/7
/// ⑦      Play  =SceneManager.LoadSceneAsync Editor dry-run=EditorSceneManager.OpenScene
/// X006      SY MOCK
/// &lt;/summary&gt;
public sealed class BootSequence
{
    public const int TotalSteps = 7;
    readonly Func&lt;bool&gt; _switchScene;
    readonly List&lt;string&gt; _lines = new();
    public BootSequence(Func&lt;bool&gt; switchScene)
    {
        _switchScene = switchScene;
    }
    public IReadOnlyList&lt;string&gt; Lines =&gt; _lines;
    public bool RunAll()
    {
        bool ok = true;
        ok &= Run(1, "Logger", LoggerStep);
        ok &= Run(2, "Time", TimeStep);
        ok &= Run(3, "Config", ConfigStep);
        ok &= Run(4, "Save", SaveStep);
        ok &= Run(5, "ILocale", LocaleStep);
        ok &= Run(6, "Ads.Preload", AdsStep);
        ok &= Run(7, "SwitchMain", () =&gt; _switchScene != null && _switchScene());
        return ok;
    }
    bool Run(int n, string name, Func&lt;bool&gt; step)
    {
        bool result;
        try
        {
            result = step();
        }
        catch (Exception e)
        {
            Logger.Error("BOOT", $"step {n} {name} exception: {e.Message}");
            result = false;
        }
        string line = $"[BOOT] {n}/{TotalSteps} {name} {(result ? "ok" : "FAIL")}";
        _lines.Add(line);
        Logger.Info("BOOT", line);
        return result;
    }
    static bool LoggerStep() =&gt; true; //      JS sink platform T001c
    static bool TimeStep()
    {
        var time = ServiceLocator.Get&lt;ITimeService&gt;();
        return time.ServerNowMs() &gt; 0 && !time.TimeCheated();
    }
}

```

```

        static bool ConfigStep()
        {
            var config = ServiceLocator.Get<IConfigService>();
            var update = config.CheckUpdate("mock://manifest");
            return update.NewVersion != null; // mock= X009 manifest
        }
        static bool SaveStep()
        {
            var save = ServiceLocator.Get<ISaveService>();
            save.Save("boot_probe", "{}");
            return !string.IsNullOrEmpty(save.Load("boot_probe"));
        }
        static bool LocaleStep()
        {
            var locale = ServiceLocator.Get<ILocaleService>();
            return !string.IsNullOrEmpty(locale.GetFontSet().NormalFont); // +
        }
        static bool AdsStep()
        {
            ServiceLocator.Get<IAdsService>().Preload("boot_warmup");
            return true;
        }
    }
}

// BootServices.cs
using System;
using System.Collections.Generic;
using System.Threading.Tasks;
using Sy.Core;
using Sy.Platform;
using UnityEngine;
using Logger = Sy.Core.Logger;
namespace Sy.P08.Boot
{
    /// <summary>
    /// Mock SY MOCK X006 Mock X002a AppID touristappid
    /// T001c WX/TT Channel Editor Boot
    /// </summary>
    public static class BootServices
    {
        public static void RegisterAll()
        {
            if (ServiceLocator.TryGet<ITimeService>(out _)) return; //
            ServiceLocator.Register<ITimeService>(new MockTimeService());
            ServiceLocator.Register<IConfigService>(new MockConfigService());
            ServiceLocator.Register<ISaveService>(new MockSaveService());
            ServiceLocator.Register<ILocaleService>(new MockLocaleService());
            ServiceLocator.Register<IAdsService>(new MockAdsService());
            Logger.Info("BOOT", "mock services registered (SY MOCK)");
        }
    }
}

```

```

}
public sealed class MockTimeService : ITimeService
{
    public TimeSource Source => TimeSource.Mock;
    public long ClientNowMs() => DateTimeOffset.UtcNow.ToUnixTimeMilliseconds();
    /// <summary>SY MOCK X006 TCB serverTime→HTTPS Date → T001b </summary>
    public long ServerNowMs() => ClientNowMs();
    public bool TimeCheated() => Math.Abs(ServerNowMs() - ClientNowMs()) > 120_000;
}
public sealed class MockConfigService : IConfigService
{
    public T GetTable<T>(string tableId) where T : class
        => throw new KeyNotFoundException("mock config table not loaded: " + tableId);
    public bool TryGetTable<T>(string tableId, out T table) where T : class
    {
        table = null;
        return false;
    }
    public ConfigUpdateResult CheckUpdate(string manifestUrl)
        => new ConfigUpdateResult(false, "mock-0.0.1");
    public void SwapPolicy(ConfigSwapPolicy policy) { }
    public string PinConfigVersion(string scope) => "mock-0.0.1";
}
public sealed class MockSaveService : ISaveService
{
    const string Prefix = "sy_save_";
    public void Save(string slot, string payloadJson) => PlayerPrefs.SetString(Prefix + slot, payloadJson);
    public string Load(string slot) => PlayerPrefs.GetString(Prefix + slot, string.Empty);
    /// <summary>X006 T001b v1→v2 </summary>
    public SyncResult CloudSync() => new SyncResult(false, null);
}
public sealed class MockLocaleService : ILocaleService
{
    public void SetLang(string lang) { }
    /// <summary>key key T001b cfg_i18n + NUnit </summary>
    public string GetText(string key, params string[] args)
    {
        Logger.Warn("LOCALE", "missing i18n key (fallback): " + key);
        return key;
    }
    public FontSet GetFontSet()
        => new FontSet("LegacyRuntime.ttf", "LegacyRuntime.ttf", "LegacyRuntime.ttf");
}
public sealed class MockAdsService : IAdsService
{
    public event Action<AdLifecycle> OnAdEvent;
    public Task<RewardResult> ShowRewarded(string pointId)
        => Task.FromResult(new RewardResult { Completed = true, Reason = "mock" });
    public void ShowInterstitial(string pointId) { }
    public void ShowBanner(string pointId) { }
}

```

```

        public void Preload(string pointId)
        {
            Logger.Info("ADS", "preload (mock): " + pointId);
            OnAdEvent?.Invoke(new AdLifecycle(AdLifecycleType.Fill, pointId, "mock"));
        }
    }
}

// MainUI.cs
using UnityEngine;
using UnityEngine.SceneManagement;
using UnityEngine.UI;
namespace Sy.P08.Newbie
{
    public static class NewbieSceneIds
    {
        public const string Main = "S08_Main";
        public const string Battle = "S08_Battle";
        public const string Explore = "S03_Explore";    // T009 case-1 exploration page
        public const string Archive = "S02_Archive";    // T009 slice-4 chapter/archive page
        public const string Unlock = "S07_Unlock";      // T015 closing slice chapter-unlock page
    }
    /// <summary>S08          →S08_Battle M5          —          </summary>
    public sealed class MainUI : MonoBehaviour
    {
        AboutPanel _aboutPanel;
        public void StartCase()
        {
            SceneManager.LoadScene(NewbieSceneIds.Battle);
        }
        /// <summary>T009 slice-2: the case-1 exploration entry (the
        /// CaseOneButton persistent listener ExploreSetup injects).
        /// T010 slice-4: every S03 entry caller names the case right
        /// before the load (the ExploreLauncher handoff contract).</summary>
        public void StartCaseOne()
        {
            Sy.P08.Adventure.ExploreLauncher.PendingCaseId = Sy.P08.Adventure.ExploreLauncher.CaseDefault;
            SceneManager.LoadScene(NewbieSceneIds.Explore);
        }
        /// <summary>T010 slice-4: the case-2 exploration entry (the
        /// CaseTwoButton persistent listener CaseTwoSetup injects) -
        /// the one S03 chain loads the case-2 nine configs.</summary>
        public void StartCaseTwo()
        {
            Sy.P08.Adventure.ExploreLauncher.PendingCaseId = "case02";
            SceneManager.LoadScene(NewbieSceneIds.Explore);
        }
        /// <summary>T009 slice-4: the chapter/archive entry (the
        /// ArchiveButton persistent listener ArchiveSetup injects).</summary>
        public void StartArchive()
        {

```

```

        SceneManager.LoadScene(NewbieSceneIds.Archive);
    }
    void Awake()
    {
        Application.targetFrameRate = 30;
        Sy.Core.MainArtInjector.Apply(transform, "P08_scene_01", "P08_unit_01");
        BuildAboutButton();
    }
    /// <summary>T030 compliance face: the about/disclosure entry on the
    /// main menu (bottom-left corner seat, programmatic UGUI - zero
    /// scene YAML, the setup-injected persistent listeners stay
    /// untouched). The label rides the cfg_about copy table.</summary>
    void BuildAboutButton()
    {
        var canvas = FindObjectOfType<Canvas>();
        if (canvas == null) return;
        _aboutPanel = AboutPanel.AttachTo(canvas.transform);
        var btnGo = new GameObject("AboutButton", typeof(RectTransform), typeof(Image), typeof(Button));
        btnGo.transform.SetParent(canvas.transform, false);
        var rt = (RectTransform)btnGo.transform;
        rt.anchorMin = rt.anchorMax = new Vector2(0f, 0f);
        rt.anchoredPosition = new Vector2(150f, 80f);
        rt.sizeDelta = new Vector2(220f, 80f);
        var img = btnGo.GetComponent<Image>();
        img.color = new Color(1f, 0.702f, 0.278f, 0.92f); // warm light #FFB347
        var labelGo = new GameObject("Label", typeof(RectTransform), typeof(Text));
        labelGo.transform.SetParent(btnGo.transform, false);
        var lrt = (RectTransform)labelGo.transform;
        lrt.anchorMin = Vector2.zero;
        lrt.anchorMax = Vector2.one;
        lrt.offsetMin = lrt.offsetMax = Vector2.zero;
        var label = labelGo.GetComponent<Text>();
        label.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
        label.text = AboutPanel.Text(AboutPanel.KeyAboutButton);
        label.fontSize = 40;
        label.alignment = TextAnchor.MiddleCenter;
        label.color = new Color(0.051f, 0.051f, 0.051f);
        label.raycastTarget = false;
        btnGo.GetComponent<Button>().onClick.AddListener(OpenAbout);
    }
    /// <summary>Opens the disclosure panel (AIGC label + ad-cap + age
    /// rating - the X433 promise-book AIGC gap closed in-game).</summary>
    void OpenAbout()
    {
        if (_aboutPanel != null) _aboutPanel.Open();
    }
}
// AchievementConfig.cs
using System;

```

```

using System.Collections.Generic;
using UnityEngine;
namespace Sy.P08.Adventure
{
    /// cfg_achievements data contract (SYS-06 / T011, the half):
    /// the achievement collection as pure data - every achievement is
    /// a goal over one tracked stat, so the whole system is
    /// config-driven ( 6 : the shipped seed covers the first
    /// six weeks of play; later batches grow the list as pure data).
    /// The folk-archive CARD half of SYS-06 stays in
    /// cfg_chapter.archiveCards (the ArchiveBook law - untouched here;
    /// the 20-card content expansion belongs to the OP02 monthly
    /// archive batches).
    /// Stat vocabulary (AchievementSystem.Stat*): verdicts_correct /
    /// ghosts_released / archive_cards / daily_solved / yinde_earned /
    /// chapters_unlocked - one stat per feed face the host already
    /// owns (VerdictSystem.VerdictCorrect / GhostReleased,
    /// ArchiveBook.CardUnlocked, DailyPuzzleSystem.DailySolved,
    /// MoralMeter.YindeGained, ChapterUnlockGate unlock callback).
    /// Plain serializable classes so JsonUtility round-trips the whole
    /// collection (config-as-data face). The achievement view slice
    /// and the P08GapCheck gate consume this one contract.
    [Serializable]
    public class AchievementDef
    {
        public string achievementId = "";
        public string title = ""; // folk title, e.g. " "
        public string desc = ""; // one-line folk copy
        public string category = ""; // / / / /
        public string statId = ""; // AchievementSystem.Stat* vocabulary
        public int goal; // >= 1 (Validate law)
    }
    [Serializable]
    public class AchievementTableFile
    {
        public string collectionId = "";
        public List<AchievementDef> achievements = new List<AchievementDef>();
    }
    /// Static load face (the AdPointTable pattern): the shipped
    /// Resources copy is the single runtime authority (the
    /// Design/configs/P08 archive copy is the versioned source; the
    /// P08GapCheck gate asserts the dual faces are byte-identical,
    /// the X364 SHA256 law).
    public static class AchievementTable
    {
        public const string ResourceName = "cfg_achievements";
        static AchievementTableFile _cache;
        public static AchievementTableFile Load(bool forceReload = false)
        {
            if (_cache != null && !forceReload) return _cache;

```

```

        var asset = Resources.Load<TextAsset>(ResourceName);
        if (asset == null)
        {
            Debug.LogError("[AchievementTable] cfg_achievements missing from Resources");
            return null;
        }
        _cache = JsonUtility.FromJson<AchievementTableFile>(asset.text);
        return _cache;
    }
}

// AchievementSystem.cs
using System;
using System.Collections.Generic;
namespace Sy.P08.Adventure
{
    /// Achievement logic core (SYS-06 / T011, the half): pure C#,
    /// zero rendering - the CaseSim pattern (the achievement view slice
    /// and the P08GapCheck gate share one rule set). The system is a
    /// stat ledger with subject-dedup faces: the host routes events
    /// the game already owns (a verdict closes, a ghost is released, a
    /// card unlocks, a daily puzzle is solved, yinde is gained, a
    /// chapter unlocks) into the matching Notify call and the ledger
    /// does the rest - no system is touched, no scene is wired, red
    /// line 12: everything is local.
    /// Stat vocabulary (cfg_achievements.statId):
    ///   verdicts_correct - correct tribunal rulings, one-shot per
    ///   caseId (re-closing a settled case never
    ///   farms the stat);
    ///   ghosts_released - EVERY wrong-verdict ghost release counts
    ///   (the M-04 beat is a performance, not a
    ///   wall - no dedup);
    ///   archive_cards - folk-archive cards unlocked, one-shot
    ///   per cardId;
    ///   daily_solved - daily mini-puzzles solved, one-shot per
    ///   dayKey;
    ///   yinde_earned - lifetime yinde gained, summed (amount
    ///   feeds must be > 0);
    ///   chapters_unlocked - chapters unlocked on any channel,
    ///   one-shot per chapterId.
    /// Laws (each has a P08GapCheck case):
    ///   A1 Validate rejects dup ids / empty copy / unknown statId /
    ///   non-positive goals before the game runs;
    ///   A2 subject one-shot: a re-notify of the same subject is
    ///   refused with zero state change (verdicts / cards / days /
    ///   chapters), ghost events always count, yinde sums;
    ///   A3 unlock law: a notify that crosses a goal unlocks the
    ///   achievement exactly once (idempotent); one notify may
    ///   unlock several achievements of the same stat;
    ///   A4 event discipline: AchievementUnlocked fires once per

```

```
///      achievement, never on refusals or below-goal notifies;
///  A5  local-authority persistence (red line 12): counters +
///      unlocked set + dedup ring round-trip through
///      CaptureState/RestoreState; a foreign collectionId or
///      corrupt json reads as a fresh ledger; unknown statIds in
///      a save are skipped silently (data-safe).
public sealed class AchievementSystem
{
    public const string StatVerdictsCorrect = "verdicts_correct";
    public const string StatGhostsReleased = "ghosts_released";
    public const string StatArchiveCards = "archive_cards";
    public const string StatDailySolved = "daily_solved";
    public const string StatYindeEarned = "yinde_earned";
    public const string StatChaptersUnlocked = "chapters_unlocked";
    /// Dedup-ring retention: the subject ring keeps this many
    /// entries so the save stays bounded (cards/chapters/cases are
    /// naturally small; the daily day-keys are the growth face -
    /// the host only notifies the CURRENT day, the ring is a dedup
    /// guard, not a calendar).
    public const int KeptSubjects = 64;
    static readonly string[] KnownStats =
    {
        StatVerdictsCorrect, StatGhostsReleased, StatArchiveCards,
        StatDailySolved, StatYindeEarned, StatChaptersUnlocked,
    };
    readonly AchievementTableFile _cfg;
    readonly Dictionary<string, int> _counters = new Dictionary<string, int>();
    readonly HashSet<string> _subjects = new HashSet<string>();
    readonly List<string> _subjectRing = new List<string>();
    readonly HashSet<string> _unlocked = new HashSet<string>();
    public event Action<AchievementDef> AchievementUnlocked;
    public AchievementSystem(AchievementTableFile cfg)
    {
        if (cfg == null) throw new ArgumentNullException("cfg");
        var problems = Validate(cfg);
        if (problems.Count > 0)
            throw new ArgumentException("achievements config invalid: " + string.Join("; ", problems.ToArray()));
        _cfg = cfg;
    }
    public AchievementTableFile Config => _cfg;
    /// The "cfg_achievements tool" half (A1): rejects any
    /// achievement data that breaks the laws before the game runs
    /// (the HintSystem.Validate pattern).
    public static List<string> Validate(AchievementTableFile cfg)
    {
        var p = new List<string>();
        if (cfg == null) { p.Add("null config"); return p; }
        if (string.IsNullOrEmpty(cfg.collectionId)) p.Add("collectionId empty");
        if (cfg.achievements == null || cfg.achievements.Count == 0)
        { p.Add("achievements empty"); return p; }
```

```

var ids = new HashSet<string>();
foreach (var a in cfg.achievements)
{
    if (a == null) { p.Add("null achievement"); continue; }
    if (string.IsNullOrEmpty(a.achievementId)) { p.Add("achievementId empty"); continue; }
    if (!ids.Add(a.achievementId)) p.Add("achievementId dup: " + a.achievementId);
    if (string.IsNullOrEmpty(a.title)) p.Add("title empty: " + a.achievementId);
    if (string.IsNullOrEmpty(a.desc)) p.Add("desc empty: " + a.achievementId);
    if (string.IsNullOrEmpty(a.category)) p.Add("category empty: " + a.achievementId);
    if (Array.IndexOf(KnownStats, a.statId) < 0)
        p.Add("statId unknown: '" + a.statId + "' @ " + a.achievementId);
    if (a.goal < 1) p.Add("goal must be >= 1: " + a.goal + " @ " + a.achievementId);
}
return p;
}
/// Stat value read (0 for unknown stats - read-only face).
public int StatValue(string statId)
{
    int v;
    return _counters.TryGetValue(statId, out v) ? v : 0;
}
public bool IsUnlocked(string achievementId)
{
    return !string.IsNullOrEmpty(achievementId) && _unlocked.Contains(achievementId);
}
public int UnlockedCount { get { return _unlocked.Count; } }
public int TotalCount { get { return _cfg.achievements.Count; } }
// — feed faces (A2): the host routes existing game events here —
/// Verdict feed: one correct ruling per caseId.
public bool NotifyVerdictCorrect(string caseId)
{
    return NotifySubject(StatVerdictsCorrect, caseId, 1);
}
/// Ghost feed: every wrong-verdict release counts (no dedup -
/// the M-04 beat is repeatable theatre).
public bool NotifyGhostReleased()
{
    return NotifyRaw(StatGhostsReleased, 1);
}
/// Archive feed: one card per cardId.
public bool NotifyArchiveCard(string cardId)
{
    return NotifySubject(StatArchiveCards, cardId, 1);
}
/// Daily feed: one solved day per dayKey.
public bool NotifyDailySolved(string dayKey)
{
    return NotifySubject(StatDailySolved, dayKey, 1);
}
/// Yinde feed: lifetime yinde gained (amount > 0).

```

```

public bool NotifyYindeGained(int amount)
{
    if (amount &lt;= 0) return false;
    return NotifyRaw(StatYindeEarned, amount);
}
/// Chapter feed: one unlock per chapterId (any channel).
public bool NotifyChapterUnlocked(string chapterId)
{
    return NotifySubject(StatChaptersUnlocked, chapterId, 1);
}
bool NotifySubject(string statId, string subject, int amount)
{
    if (string.IsNullOrEmpty(subject)) return false;
    string key = statId + "#" + subject;
    if (!_subjects.Contains(key)) return false;          // A2 one-shot
    _subjects.Add(key);
    _subjectRing.Add(key);
    while (_subjectRing.Count &gt; KeptSubjects)
    {
        var dropped = _subjectRing[0];
        _subjectRing.RemoveAt(0);
        _subjects.Remove(dropped);
    }
    return NotifyRaw(statId, amount);
}
bool NotifyRaw(string statId, int amount)
{
    int v;
    _counters.TryGetValue(statId, out v);
    _counters[statId] = v + amount;
    CheckUnlocks(statId);
    return true;
}
/// A3 unlock law: every achievement of this stat at or over its
/// goal unlocks exactly once. Idempotent by the unlocked set.
void CheckUnlocks(string statId)
{
    int v = StatValue(statId);
    foreach (var a in _cfg.achievements)
    {
        if (a == null || a.statId != statId) continue;
        if (v &lt; a.goal) continue;
        if (!_unlocked.Add(a.achievementId)) continue;
        AchievementUnlocked?.Invoke(a);
    }
}
// — local-authority persistence (A5, red line 12) —
[Serializable]
public class StatPair
{

```

```

        public string statId = "";
        public int value;
    }
    [Serializable]
    public class SaveState
    {
        public string collectionId = "";
        public List<StatPair> counters = new List<StatPair>();
        public List<string> subjects = new List<string>();
        public List<string> unlocked = new List<string>();
    }
    /// Captures the ledger as a JsonUtility string. The HOST owns
    /// persistence (PlayerPrefs local slot) - the core stays pure.
    public string CaptureState()
    {
        var s = new SaveState { collectionId = _cfg.collectionId };
        foreach (var kv in _counters)
            s.counters.Add(new StatPair { statId = kv.Key, value = kv.Value });
        foreach (var sub in _subjectRing) s.subjects.Add(sub);
        foreach (var id in _unlocked) s.unlocked.Add(id);
        return UnityEngine.JsonUtility.ToJson(s);
    }
    /// Restores the ledger. Returns "" on success or the rejection
    /// reason (A5: the caller falls back to a fresh ledger - a bad
    /// save never wipes earned achievements). Unknown statIds and
    /// foreign achievement ids in the save are skipped silently
    /// (data-safe); restore never fires AchievementUnlocked
    /// (resume must not replay unlock toasts).
    public string RestoreState(string json)
    {
        if (string.IsNullOrEmpty(json)) return "empty save";
        SaveState s;
        try { s = UnityEngine.JsonUtility.FromJson<SaveState>(json); }
        catch (Exception e) { return "bad json: " + e.Message; }
        if (s == null) return "null save";
        if (s.collectionId != _cfg.collectionId) return "collection id mismatch: " + s.collectionId;
        var authored = new HashSet<string>();
        foreach (var a in _cfg.achievements)
            if (a != null) authored.Add(a.achievementId);
        _counters.Clear();
        if (s.counters != null)
            foreach (var pair in s.counters)
            {
                if (pair == null || Array.IndexOf(KnownStats, pair.statId) < 0) continue;
                if (pair.value < 0) return "negative counter: " + pair.statId;
                _counters[pair.statId] = pair.value;
            }
        _subjects.Clear();
        _subjectRing.Clear();
        if (s.subjects != null)

```

```

        foreach (var sub in s.subjects)
        {
            if (string.IsNullOrEmpty(sub) || !_subjects.Contains(sub)) continue;
            _subjects.Add(sub);
            _subjectRing.Add(sub);
        }
        while (_subjectRing.Count > KeptSubjects)
        {
            var dropped = _subjectRing[0];
            _subjectRing.RemoveAt(0);
            _subjects.Remove(dropped);
        }
        _unlocked.Clear();
        if (s.unlocked != null)
            foreach (var id in s.unlocked)
                if (!string.IsNullOrEmpty(id) && authored.Contains(id)) _unlocked.Add(id);
        return "";
    }
    /// Test hygiene (house rule): resets the whole ledger so a
    /// check run never inherits another case's progress.
    public void ClearForTests()
    {
        _counters.Clear();
        _subjects.Clear();
        _subjectRing.Clear();
        _unlocked.Clear();
    }
}

// AdPointConfig.cs
using System;
using UnityEngine;
namespace Sy.P08.Adventure
{
    /// cfg_ad_point data contract (T015): the chapter-economy ad point
    /// table - the task book section four V1.1 final values as pure data.
    /// Plain serializable classes so JsonUtility round-trips the whole
    /// funnel table (config-as-data face). The AdPointFunnel static face
    /// and the AdFunnelCheck gate consume this one contract - contract
    /// before data, no orphan data.
    [Serializable]
    public class AdPointTableFile
    {
        public int globalCapDaily = 12;
        public float[] gainDecayLadder = new float[0];
        public AdPriorityLadderBlock priorityLadder;
        public AdPointRow[] points = new AdPointRow[0];
    }
    [Serializable]
    public class AdPriorityLadderBlock

```

```

{
    public int rescue;
    public int @double;    // C# keyword escape; JsonUtility maps the JSON "double" key
    public int speedup;
    public int direct;
    public int misc;
}
[Serializable]
public class AdGateBlock
{
    public int channels;          // chapter_unlock: 2 ads / 2 days / yinde-free
    public int adViewsPerUnlock;
    public string freePass = "";
    public int yindeFreeCost;     // chapter_unlock: S07 yinde channel seed cost (declared assumption in cfg meta;
    public bool chapterEndOnly;   // interstitial: sim-side chapter-end gate
}
[Serializable]
public class AdPointRow
{
    public string pointId = "";
    public string type = "";      // "rewarded" | "interstitial"
    public string @class = "";   // rescue / double / speedup / direct / misc
    public int priority;
    public int dailyCap;          // design cap, dimension = capScope (sim-side)
    public string capScope = ""; // "puzzle" | "chapter"
    public int dayCap;            // AdCapPolicy daily safety-valve pool
    public string adCategory = ""; // p08_-isolated category id (AdCapPolicy wiring)
    public int coolDownSec;
    public AdGateBlock gate;      // null when the row carries no gate block
    public string decay = "";     // empty = constant full value; "gainLadder" = decay face
    public string rewardRef = "";
    public string uiStyle = "";
}
/// Static load face: the shipped Resources copy is the single runtime
/// authority (the Design/configs/P08 archive copy is the versioned
/// source; the AdFunnelCheck gate asserts the dual faces are
/// byte-identical, the X364 SHA256 law).
public static class AdPointTable
{
    static AdPointTableFile _cache;
    public static AdPointTableFile Load(bool forceReload = false)
    {
        if (_cache != null && !forceReload) return _cache;
        var asset = Resources.Load<TextAsset>("cfg_ad_point");
        if (asset == null)
        {
            Debug.LogError("[AdPointTable] cfg_ad_point missing from Resources");
            return null;
        }
        _cache = JsonUtility.FromJson<AdPointTableFile>(asset.text);
    }
}

```

```

        return _cache;
    }
    /// Table row for a cfg point id (null = unknown id).
    public static AdPointRow Row(string pointId)
    {
        var table = Load();
        if (table?.points == null) return null;
        foreach (var p in table.points)
            if (p != null && p.pointId == pointId) return p;
        return null;
    }
}
}
// AdPointFunnel.cs
using System.Collections.Generic;
using Sy.Platform;
using Sy.Telemetry;
namespace Sy.P08.Adventure
{
    /// T015 - the cfg_ad_point consumer face: the P02 three-piece pattern
    /// (HasCategoryBudget pre-check -& category-gated consume -& evt_ad
    /// funnel telemetry) as pure functions over the shared AdCapPolicy, so
    /// the S07 chapter-lock view, the wen-mi host (HintSystem), the archive
    /// bonus face and the AdFunnelCheck gate stay thin call sites.
    /// Per-puzzle and per-chapter caps stay sim-side (HintSystem W1
    /// tier==ads ladder, ArchiveBook SYS-06 one-ad-card law, the chapter
    /// lock's 2-ads-or-2-days-or-yinde-free channels); this face owns the
    /// day pools and the interstitial face. Zero server (red line 12) -
    /// policy and table are both local; a missing table row reads as
    /// "no offer" (fail-loud callers surface the missing asset through
    /// AdPointTable.Load's error log).
    public static class AdPointFunnel
    {
        /// Runtime ad point id: p08_-isolated namespace (P01 T020
        /// precedent) - the ads service and evt_ad telemetry never see a
        /// bare id that could collide with the P01/P02/P06/G namespaces.
        public static string AdPointId(string pointId) => "p08_" + pointId;
        /// Table row for a cfg point id (null = unknown id).
        public static AdPointRow Row(string pointId) => AdPointTable.Row(pointId);
        /// The shared policy wired exactly as the table reads: the global
        /// hard cap, the interstitial cooldown and the interstitial day
        /// valve all come from the data table (single authority).
        public static AdCapPolicy CreatePolicy(AdPointTableFile table)
        {
            int global = table != null && table.globalCapDaily > 0
                ? table.globalCapDaily : AdCapPolicy.DefaultDailyCap;
            float cool = AdCapPolicy.DefaultInterstitialCooldownSec;
            int interCap = 0;
            if (table != null && table.points != null)
                foreach (var p in table.points)

```

```

        {
            if (p == null || p.type != "interstitial") continue;
            if (p.coolDownSec > 0) cool = p.coolDownSec;
            if (p.dayCap > 0) interCap = p.dayCap;
        }
        return new AdCapPolicy(global, cool, interCap);
    }

    /// Section-six pre-offer gate (P02 paradigm): true when the point's
    /// p08_ category still has day budget. The global cap is enforced at
    /// consume; per-puzzle/per-chapter caps are sim-side and owned by
    /// the host.
    public static bool PrecheckBudget(string pointId, string dayKey, AdCapPolicy policy)
    {
        var row = Row(pointId);
        if (row == null || row.type != "rewarded") return false;
        return policy.HasCategoryBudget(dayKey, AdCategories.CategoryOfPoint(AdPointId(pointId)));
    }

    /// Category-gated consume (P02 paradigm): day pool first, then the
    /// global hard cap; both counters advance together. False = the
    /// host tracks the funnel "blocked" state and never retries the ad.
    public static bool TryConsume(string pointId, string dayKey, AdCapPolicy policy)
    {
        var row = Row(pointId);
        if (row == null || row.type != "rewarded") return false;
        return policy.TryConsumeRewarded(dayKey, AdCategories.CategoryOfPoint(AdPointId(pointId)));
    }

    /// Gain decay face (yinde_boost only): gain-class points walk the
    /// ladder by same-day watch index (watch 1 = 1.0, 2 = 0.9, ...);
    /// every other P08 point keeps full value on every watch. watchIdx
    /// is 1-based; a missing table reads as full value.
    public static float GainMul(string pointId, int watchIdx)
    {
        var row = Row(pointId);
        var ladder = AdPointTable.Load()?.gainDecayLadder;
        if (row == null || ladder == null || ladder.Length == 0) return 1f;
        if (string.IsNullOrEmpty(row.decay)) return 1f;
        int idx = watchIdx <= 1 ? 0
            : watchIdx > ladder.Length ? ladder.Length - 1
            : watchIdx - 1;
        return ladder[idx];
    }

    /// Interstitial face: chapter-end only (the sim-side gate - the
    /// host never offers mid-chapter) riding the policy's cooldown +
    /// day valve (the most restrained interstitial in the portfolio).
    public static bool InterstitialAllowed(bool chapterEnd, string dayKey, float nowSec, AdCapPolicy policy)
    {
        if (!chapterEnd) return false;
        if (Row("interstitial") == null) return false;
        return policy.TryConsumeInterstitial(dayKey, nowSec);
    }
}

```

```

    /// evt_ad funnel telemetry (P02 vocabulary): states =
    /// offer/accept/decline/ad_fail/blocked/complete. The emitted
    /// pointId is the p08_-isolated runtime id; chapterId rides the
    /// fields so the chapter P&L face can read per-chapter funnels
    /// from the local log (zero server, red line 12).
    public static void Track(TelemetryService telemetry, string state,
        string pointId, string chapterId, string reason)
    {
        if (telemetry == null) return;
        var fields = new Dictionary<string, string>
        {
            { "state", state },
            { "pointId", AdPointId(pointId) },
        };
        if (chapterId != null) fields["chapterId"] = chapterId;
        if (reason != null) fields["reason"] = reason;
        telemetry.Track(EventIds.Ad, fields);
    }
}

// ArchiveBook.cs
using System;
using System.Collections.Generic;
namespace Sy.P08.Adventure
{
    /// Folk archive book logic core (T009): pure C#, zero rendering -
    /// the CaseSim pattern. One book owns one chapter's collectible
    /// knowledge cards; the chapter view slice, the rewarded-ad host
    /// and the ArchiveCheck gate share this one rule set.
    /// Laws:
    /// A1 clue-pinned cards unlock exactly when their board clue is
    /// found - repeated finds and unknown clue ids change nothing;
    /// A2 the chapter's single archive_extra card unlocks only via
    /// the rewarded-ad callback (GrantAdUnlock) - never via clues,
    /// and clue cards never unlock via the callback;
    /// A3 at most one ad card per chapter (SYS-06 archive_extra law)
    /// - a grant past the cap is refused with zero state change;
    /// A4 local-authority break resume: CaptureState/RestoreState
    /// round-trips the unlocked set silently (no events on
    /// restore); a state from another chapter is refused and the
    /// book stays fresh (red line 12: zero server, bad save data
    /// never blocks play).
    public sealed class ArchiveBook
    {
        public const int MaxAdCardsPerChapter = 1; // SYS-06 archive_extra law
        readonly ChapterConfig _chapter;
        readonly HashSet<string> _unlocked = new HashSet<string>();
        readonly Dictionary<string, List<ArchiveCardDef>>> _cluePins =
            new Dictionary<string, List<ArchiveCardDef>>>(); // clueId -> pinned cards
        ArchiveCardDef _adCard; // null when the chapter ships none
    }
}

```

```
public ArchiveBook(ChapterConfig chapter)
{
    if (chapter == null) throw new ArgumentNullException("chapter");
    var problems = Validate(chapter);
    if (problems.Count > 0)
        throw new ArgumentException("chapter config invalid: " + string.Join("; ", problems.ToArray()));
    _chapter = chapter;
    foreach (var card in chapter.archiveCards)
    {
        if (card.unlockType == "ad") { _adCard = card; continue; }
        List<ArchiveCardDef> pins;
        if (!_cluePins.TryGetValue(card.unlockRef, out pins))
        {
            pins = new List<ArchiveCardDef>();
            _cluePins[card.unlockRef] = pins;
        }
        pins.Add(card);
    }
}

public event Action<string> CardUnlocked;    // (cardId) - fires on real unlocks only
/// The "cfg_chapter tool" half: rejects any chapter data that
/// breaks the archive laws before the game runs (the
/// HintSystem.Validate pattern). Returns the problem list
/// (empty = valid).
public static List<string> Validate(ChapterConfig c)
{
    var p = new List<string>();
    if (c == null) { p.Add("null chapter"); return p; }
    if (string.IsNullOrEmpty(c.chapterId)) p.Add("chapterId empty");
    if (c.scenes == null || c.scenes.Count == 0) p.Add("scenes empty");
    else
    {
        var seen = new HashSet<string>();
        foreach (var s in c.scenes)
        {
            if (string.IsNullOrEmpty(s)) { p.Add("scene id empty"); continue; }
            if (!seen.Add(s)) p.Add("scene id dup: " + s);
        }
    }
    if (c.puzzles == null) p.Add("puzzles null");
    else
    {
        var seen = new HashSet<string>();
        foreach (var pz in c.puzzles)
        {
            if (string.IsNullOrEmpty(pz)) { p.Add("puzzle id empty"); continue; }
            if (!seen.Add(pz)) p.Add("puzzle id dup: " + pz);
        }
    }
    if (string.IsNullOrEmpty(c.verdict)) p.Add("verdict empty");
}
```

```

    if (string.IsNullOrEmpty(c.freePass)) p.Add("freePass empty");
    if (c.archiveCards == null || c.archiveCards.Count == 0) p.Add("archiveCards empty");
    else
    {
        int adCards = 0;
        var ids = new HashSet<string>();
        foreach (var card in c.archiveCards)
        {
            if (card == null) { p.Add("null card"); continue; }
            if (string.IsNullOrEmpty(card.cardId)) { p.Add("cardId empty"); continue; }
            if (!ids.Add(card.cardId)) p.Add("cardId dup: " + card.cardId);
            if (string.IsNullOrEmpty(card.title)) p.Add("card title empty: " + card.cardId);
            if (string.IsNullOrEmpty(card.category)) p.Add("card category empty: " + card.cardId);
            if (string.IsNullOrEmpty(card.body)) p.Add("card body empty: " + card.cardId);
            if (card.unlockType == "clue")
            {
                if (string.IsNullOrEmpty(card.unlockRef)) p.Add("clue card ref empty: " + card.cardId);
            }
            else if (card.unlockType == "ad")
            {
                if (!string.IsNullOrEmpty(card.unlockRef)) p.Add("ad card ref must be empty: " + card.cardId);
                adCards++;
            }
            else p.Add("unlockType unknown: '" + card.unlockType + "' @ " + card.cardId);
        }
        if (adCards > MaxAdCardsPerChapter) p.Add("ad cards over cap: " + adCards);
    }
    return p;
}

public ChapterConfig Chapter { get { return _chapter; } }
public IList<ArchiveCardDef> Cards { get { return _chapter.archiveCards; } }
public int UnlockedCount { get { return _unlocked.Count; } }
public int TotalCount { get { return _chapter.archiveCards.Count; } }
public bool IsUnlocked(string cardId)
{
    return !string.IsNullOrEmpty(cardId) && _unlocked.Contains(cardId);
}

/// The board-clue face (A1): the host calls this when a clue
/// lands on the verdict board. Unlocks every card pinned to
/// that clue; returns whether anything newly unlocked. Unknown
/// ids and repeated finds change nothing.
public bool NotifyClueFound(string clueId)
{
    if (string.IsNullOrEmpty(clueId)) return false;
    List<ArchiveCardDef> pins;
    if (!_cluePins.TryGetValue(clueId, out pins)) return false;
    bool any = false;
    foreach (var card in pins)
    {
        if (_unlocked.Add(card.cardId))
    }

```

```
        {
            any = true;
            CardUnlocked?.Invoke(card.cardId);
        }
    }
    return any;
}

/// The archive_extra rewarded-ad callback (A2/A3): unlocks the
/// chapter's single ad card. Refused (zero state change) when
/// the chapter ships no ad card or it is already unlocked. The
/// HOST owns the ad SDK face - this core never touches it
/// (red line 12).
public bool GrantAdUnlock()
{
    if (_adCard == null) return false;
    if (!_unlocked.Add(_adCard.cardId)) return false;
    CardUnlocked?.Invoke(_adCard.cardId);
    return true;
}

/// Local-authority break resume (A4): capture the unlocked set
/// for the host's save file.
[Serializable]
public class ArchiveBookState
{
    public string chapterId = "";
    public List<string> unlocked = new List<string>();
}

public ArchiveBookState CaptureState()
{
    var st = new ArchiveBookState();
    st.chapterId = _chapter.chapterId;
    foreach (var id in _unlocked) st.unlocked.Add(id);
    return st;
}

/// Restore into this book: refuses foreign-chapter states; ids
/// this chapter never authored are skipped (data-safe). Restore
/// is silent - it never fires CardUnlocked (resume must not
/// replay unlock toasts).
public bool RestoreState(ArchiveBookState st)
{
    if (st == null || st.chapterId != _chapter.chapterId) return false;
    var authored = new HashSet<string>();
    foreach (var card in _chapter.archiveCards) authored.Add(card.cardId);
    foreach (var id in st.unlocked)
        if (!string.IsNullOrEmpty(id) && authored.Contains(id)) _unlocked.Add(id);
    return true;
}
}

// ArchiveLauncher.cs
```

```
using UnityEngine;
using UnityEngine.SceneManagement;
namespace Sy.P08.Adventure
{
    /// <summary>
    /// T009 slice-4 host for the S02 chapter/archive page (T010
    /// slice-4: case-parameterized via the PendingCaseId handoff):
    /// loads the shipped configs from Resources (the runtime double
    /// face, byte identical to Design/configs/P08 - X364 law),
    /// assembles ArchiveBook + ArchiveView and routes the view's input
    /// intents: the shipped chapter tap into the real S03 scene load
    /// (the S02 -> S03 flow, requirement doc section 3), the
    /// archive_extra seat into ArchiveBook.GrantAdUnlock (the host owns
    /// the rewarded-ad callback face - the core never touches the SDK,
    /// red line 12) and the deterministic case-code share seat (the
    /// X372 promise: the settled-case code lands on this page). The
    /// morality seat shows the settled newbie baseline; the T012
    /// MoralMeter owns the live source later. Archive unlock
    /// persistence (ArchiveBook A4 capture/restore) stays core-asserted
    /// this slice - the host save wiring lands together with the
    /// chapter-progress save face (T011/T012 era), so the book and the
    /// clue board always save as one unit, never out of sync.
    /// </summary>
    public sealed class ArchiveLauncher : MonoBehaviour
    {
        ArchiveBook _book;
        ArchiveView _view;
        ChapterUnlockGate _gate;    // T015 closing slice: owns the chapter-row routing
        string _secondChapterId;    // probed from cfg_chapter_case02 (null until it ships)
        string _secondFreePass;    // the second chapter's own free-pass window
        void Awake()
        {
            Application.targetFrameRate = 30;
            // T010 slice-4: case-parameterized book face (the
            // PendingCaseId static handoff, the ExploreLauncher
            // pattern) - the page opens the pending case's archive;
            // the default keeps case-1 behaviour bit-identical.
            string caseId = ExploreLauncher.PendingCaseId;
            var chapter = Parse<ChapterConfig>("cfg_chapter_" + caseId);
            var verdict = Parse<VerdictConfig>("cfg_verdict_" + caseId);
            _book = new ArchiveBook(chapter);
            var canvas = FindObjectOfType<Canvas>();
            _view = gameObject.AddComponent<ArchiveView>();
            _view.Initialize(canvas, _book);
            // T010 slice-4: the chapter row addresses shipped cases by
            // id - the second seat flips live exactly when
            // cfg_chapter_case02 ships in Resources (data-driven probe;
            // a missing config keeps the locked dead face).
            var secondAsset = Resources.Load<TextAsset>("cfg_chapter_case02");
            _secondChapterId = null;
        }
    }
}
```

```
_secondFreePass = "";
if (secondAsset != null)
{
    var second = JsonUtility.FromJson<ChapterConfig>(secondAsset.text);
    if (second != null)
    {
        _secondChapterId = second.chapterId;
        _secondFreePass = second.freePass;
    }
}
_view.SetChapterRow(ExploreLauncher.CaseDefault, _secondChapterId, secondAsset != null);
// T015 closing slice: the shared three-channel gate owns the
// chapter-row routing (ad/days/yinde channels, launch free-pass).
_gate = new ChapterUnlockGate(
    AdPointFunnel.CreatePolicy(AdPointTable.Load(true)),
    new Sy.Telemetry.TelemetryService());
// Host-fed seats: the case code recomputes locally from the
// shipped config (any device gets the same code - red line 12).
_view.SetMorality(10); // settled newbie baseline (Redline11Check A8)
_view.SetCaseCode(" " + FinalView.CaseCode(verdict));
_view.ChapterTapped += OnChapterTapped;
_view.AdTapped += OnAdTapped;
_view.SyncView();
}
void Update()
{
    if (_view == null) return;
    _view.SyncView();
}
void OnChapterTapped(string chapterId)
{
    // S02 -&gt; S03 flow (T015 closing slice): only shipped ids ever
    // reach this wire. The gate owns the routing - a free-pass or
    // unlocked chapter loads the one S03 chain exactly as before
    // (zero regression for the launch window); a locked chapter
    // diverts to the S07 unlock page with the static handoff.
    if (chapterId != ExploreLauncher.CaseDefault && chapterId != _secondChapterId) return;
    int index = chapterId == ExploreLauncher.CaseDefault ? 1 : 2;
    string freePass = chapterId == ExploreLauncher.CaseDefault
        ? _book.Chapter.freePass : _secondFreePass;
    if (RouteFor(_gate, chapterId, index, freePass, UnlockLauncher.TodayKey()) == RouteUnlock)
    {
        UnlockLauncher.PendingUnlockId = chapterId;
        SceneManager.LoadScene(Sy.P08.Newbie.NewbieSceneIds.Unlock);
        return;
    }
    ExploreLauncher.PendingCaseId = chapterId;
    SceneManager.LoadScene(Sy.P08.Newbie.NewbieSceneIds.Explore);
}
// Routing decision (pure, UnlockPageCheck-asserted): a locked
```

```

// chapter diverts to the S07 unlock page; free-pass and unlocked
// chapters keep the pre-gate behaviour (zero regression).
public const string RouteExplore = "explore";
public const string RouteUnlock = "unlock";
public static string RouteFor(ChapterUnlockGate gate, string chapterId, int chapterIndex, string freePass, string
{
    var face = gate.Evaluate(chapterId, chapterIndex, freePass, todayKey);
    return face.status == ChapterUnlockGate.StatusLocked ? RouteUnlock : RouteExplore;
}
void OnAdTapped()
{
    // The host owns the rewarded-ad face: the grant stands in for
    // the ad callback (A2 single channel, the core never touches
    // the SDK).
    _book.GrantAdUnlock();
}
static T Parse<T>(string resource) where T : class
{
    var asset = Resources.Load<TextAsset>(resource);
    if (asset == null) throw new System.Exception("runtime config missing: " + resource);
    var cfg = JsonUtility.FromJson<T>(asset.text);
    if (cfg == null) throw new System.Exception("runtime config parse failed: " + resource);
    return cfg;
}
}
}
// ArchiveView.cs
using System;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.UI;
namespace Sy.P08.Adventure
{
    /// <summary>
    /// T009 slice-4: the S02 chapter/archive view (polished greybox;
    /// T010 slice-4: the chapter row is host-fed and case-addressed -
    /// a shipped second case flips its seat live, a not-yet-shipped
    /// one stays a locked dead face). Pure programmatic UGUI in the
    /// FinalView pattern - every element is built in Initialize (zero
    /// scene-YAML drift). Faces:
    /// the chapter-select row (shipped chapter live, future chapter
    /// locked with its free-pass window note), the folk archive book
    /// (the five cards from cfg_chapter - unlocked cards show their
    /// verbatim title/category, locked cards stay dark), the single
    /// archive_extra rewarded-ad seat (live only while the ad card is
    /// still locked - the HOST owns the ad callback face, red line 12),
    /// the case-code share seat (the X372 promise: the settled case
    /// code lands here, host-fed, deterministic - any device recomputes
    /// it from the shipped config) and the morality seat (host-fed;
    /// the T012 MoralMeter owns the live source later). Tapping an

```

```

/// unlocked card opens the aged-paper reading overlay - the body
/// is typeset through the T004 VerticalTextLayout solver (pure
/// transform, V1-V10 laws live in this face). Game text comes
/// from the configs verbatim - the view hardcodes no case content.
/// &lt;/summary&gt;
public sealed class ArchiveView : MonoBehaviour
{
    Canvas _canvas;
    Sprite _square;
    ArchiveBook _book;
    Text _titleText, _toastText, _moralityText, _codeText, _counterText, _freePassText;
    Button _chapterOneButton, _chapterTwoButton, _adButton;
    RectTransform _detailOverlay;
    Text _detailTitle, _detailCategory;
    readonly Image[] _cardFills;
    readonly Text[] _cardTexts;
    readonly Button[] _cardButtons;
    readonly Text[] _cardSeals;           // T021 red seal stamp on unlocked cards
    readonly Text[] _cardCats;           // T021 slice-2 deep layout: category line under the title
    readonly Image[] _cardChips;         // T021 slice-2: category glyph chip at the left edge
    readonly string[] _cardIds;
    string _detailBodyRaw = "";
    int _detailColumns;
    // T010 slice-4: chapter-row ids (host-fed). Defaults keep the
    // pre-slice mount bit-identical when the host never calls
    // SetChapterRow - the book's own chapter id stays the first id
    // and the second seat stays the locked dead face.
    string _chapterOneId;
    string _chapterTwoId;
    bool _chapterTwoLive;
    public event Action<string> ChapterTapped; // (chapterId) -&gt; host loads the chapter scene
    public event Action AdTapped;             // -&gt; host calls book.GrantAdUnlock
    // Art spec sheet palette (StyleGateCheck B1 locks the same hex).
    static readonly Color CurtainRed = new Color(0.545f, 0f, 0f);           // #8B0000
    static readonly Color WarmYellow = new Color(1f, 0.702f, 0.278f);       // #FFB347
    static readonly Color SilhouetteBlack = new Color(0.051f, 0.051f, 0.051f); // #0D0D0D
    static readonly Color PaperWhite = new Color(0.961f, 0.902f, 0.784f);    // #F5E6C8
    static readonly Color NetherBlue = new Color(0.106f, 0.227f, 0.294f);    // #1B3A4B
    static readonly Color TalismanGold = new Color(1f, 0.839f, 0.376f);      // #FFD660
    // Derived (MUST stay textually after their sources, pitfall #119).
    static readonly Color SlotDark = new Color(0.12f, 0.10f, 0.08f, 0.9f);
    static readonly Color ChapterLocked = new Color(0.10f, 0.08f, 0.07f, 0.85f);
    const float SlotTop = 150f;      // first card slot center Y
    const float SlotGap = 170f;      // slot pitch
    const int BodyCharsPerColumn = 16; // T004 vertical typesetting height
    const string LockedLabel = "      ";
    public ArchiveView()
    {
        int n = 8; // ceiling: cfg_chapter_case01 ships five cards; extra slots stay hidden
        _cardFills = new Image[n];

```

```
_cardTexts = new Text[n];
_cardButtons = new Button[n];
_cardSeals = new Text[n];
_cardCats = new Text[n];
_cardChips = new Image[n];
_cardIds = new string[n];
}

public void Initialize(Canvas canvas, ArchiveBook book)
{
    _canvas = canvas;
    _book = book;
    _chapterOneId = book.Chapter.chapterId;
    _square = MakeSquareSprite();
    BuildStaticFace();
    _book.CardUnlocked += OnCardUnlocked;
    SyncView();
}

// —— input intents (host routes them into the cores) ——
/// &summary>The gate face of a chapter tap: same path the Button
/// walks, so the navigation the player uses is what gets
/// asserted. The locked future chapter carries NO listener -
/// its button is a dead face by construction.&summary>
public void SimulateChapterTap(string chapterId)
{
    // T010 slice-4: ids route to their own seat; unknown or
    // locked ids are dead faces (no navigation, zero intents).
    if (chapterId == _chapterOneId)
    {
        if (_chapterOneButton != null) _chapterOneButton.onClick.Invoke();
    }
    else if (_chapterTwoLive && chapterId == _chapterTwoId)
    {
        if (_chapterTwoButton != null) _chapterTwoButton.onClick.Invoke();
    }
}

public void SimulateCardTap(string cardId)
{
    for (int i = 0; i < _cardIds.Length; i++)
        if (_cardIds[i] == cardId && _cardButtons[i] != null)
        {
            _cardButtons[i].onClick.Invoke();
            return;
        }
}

public void SimulateAdTap()
{
    if (_adButton != null && _adButton.gameObject.activeSelf) _adButton.onClick.Invoke();
}

/// &summary>Tap-to-close on the reading overlay: same path the
/// overlay Button walks.&summary>
```

```

public void SimulateDetailTap()
{
    var btn = _detailOverlay != null ? _detailOverlay.GetComponent<Button>() : null;
    if (btn != null && _detailOverlay.gameObject.activeSelf) btn.onClick.Invoke();
}
// —— core event handlers ——
void OnCardUnlocked(string cardId)
{
    SetToast(string.IsNullOrEmpty(HasCard(cardId)) ? "    " : "    " + CardTitle(cardId) + " ");
    SyncView();
}
// —— host-facing setters (launcher parity) ——
/// <summary>T010 slice-4: the host feeds the chapter row - both
/// shipped cases ride the one S03 chain, so the seats address
/// cases by id. The second seat flips live only when the host
/// probed its chapter config in Resources (data-driven); until
/// then it stays the locked dead face with no listener. Called
/// after Initialize; unknown ids never route.</summary>
public void SetChapterRow(string firstId, string secondId, bool secondLive)
{
    if (!string.IsNullOrEmpty(firstId)) _chapterOneId = firstId;
    _chapterTwoId = secondId;
    if (_chapterTwoButton == null) return;
    _chapterTwoLive = secondLive && !string.IsNullOrEmpty(secondId);
    _chapterTwoButton.interactable = _chapterTwoLive;
    _chapterTwoButton.image.color = _chapterTwoLive ? NetherBlue : ChapterLocked;
    var label = _chapterTwoButton.GetComponentInChildren<Text>();
    if (label != null)
        label.text = _chapterTwoLive ? "    " : "    ";
    _chapterTwoButton.onClick.RemoveAllListeners();
    if (_chapterTwoLive)
        _chapterTwoButton.onClick.AddListener(() => ChapterTapped?.Invoke(_chapterTwoId));
}
public void SetToast(string text) { if (_toastText != null) _toastText.text = text; }
/// <summary>The morality seat (S02 law): the host feeds the value;
/// the T012 MoralMeter owns the live source later.</summary>
public void SetMorality(int value) { if (_moralityText != null) _moralityText.text = "    " + value; }
/// <summary>The case-code share seat (X372 promise): the host feeds
/// the deterministic code string.</summary>
public void SetCaseCode(string text) { if (_codeText != null) _codeText.text = text; }
// —— gate/state queries (ChapterGateCheck face) ——
public int ShippedCardCount { get { return _book != null ? _book.TotalCount : 0; } }
public bool IsAdSeatLive
{
    get { return _adButton != null && _adButton.gameObject.activeSelf; }
}
public bool IsDetailOpen { get { return _detailOverlay != null && _detailOverlay.gameObject.activeSelf; } }
public string DetailTitleText { get { return _detailTitle != null ? _detailTitle.text : ""; } }
public string DetailCategoryText { get { return _detailCategory != null ? _detailCategory.text : ""; } }
/// <summary>The raw body the reading overlay carries - asserted

```

```

    /// verbatim against the config (game text = cfg law).&lt;/summary&gt;
    public string DetailBodyText { get { return _detailBodyRaw; } }
    /// &lt;summary&gt;How many vertical columns the T004 solver produced for
    /// the open card (0 when no detail is open).&lt;/summary&gt;
    public int DetailColumnCount { get { return _detailColumns; } }
    public string MoralityText { get { return _moralityText != null ? _moralityText.text : ""; } }
    public string CaseCodeText { get { return _codeText != null ? _codeText.text : ""; } }
    public string CounterText { get { return _counterText != null ? _counterText.text : ""; } }
    public string ToastText { get { return _toastText != null ? _toastText.text : ""; } }
    public bool IsChapterOneLive
    {
        get { return _chapterOneButton != null && _chapterOneButton.interactable; }
    }
    public bool IsChapterTwoLocked
    {
        get { return _chapterTwoButton != null && !_chapterTwoButton.interactable; }
    }
    /// &lt;summary&gt;T010 slice-4: the second seat's live face (true only
    /// after the host feeds a shipped second chapter id).&lt;/summary&gt;
    public bool IsChapterTwoLive
    {
        get { return _chapterTwoButton != null && _chapterTwoButton.interactable; }
    }
    public Color SlotColor(int index)
    {
        return index &gt;= 0 && index &lt; _cardFills.Length && _cardFills[index] != null
            ? _cardFills[index].color
            : Color.clear;
    }
    public string SlotLabel(int index)
    {
        if (index &lt; 0 || index &gt;= _cardTexts.Length || _cardTexts[index] == null) return "";
        // Gate contract kept verbatim: the unlocked slot still reads
        // "title | category" even though the face now typesets the two
        // on separate lines (deep layout slice 2).
        if (_book != null && _cardCats[index] != null && !string.IsNullOrEmpty(_cardIds[index])
            && _book.IsUnlocked(_cardIds[index]))
            return _cardTexts[index].text + "  " + _cardCats[index].text;
        return _cardTexts[index].text;
    }
    // —— card face (tap = the reading law) ——
    void OnCardButtonTapped(int index)
    {
        string cardId = _cardIds[index];
        if (!_book.IsUnlocked(cardId))
        {
            SetToast("    —    ");
            return;
        }
        OpenReadingFace(cardId);
    }

```

```
}
void OpenReadingFace(string cardId)
{
    if (string.IsNullOrEmpty(HasCard(cardId))) return;
    var card = CardOf(cardId);
    _detailTitle.text = card.title;
    _detailCategory.text = card.category;
    _detailBodyRaw = card.body;
    // T004 vertical typesetting (pure transform - V1..V10 laws):
    // columns render right-to-left, folk scroll reading order.
    var columns = VerticalTextLayout.Layout(card.body, BodyCharsPerColumn);
    _detailColumns = columns.Count;
    var leftToRight = VerticalTextLayout.RenderRightToLeft(columns);
    float colW = 72f, bodyW = leftToRight.Count * colW;
    float x0 = -bodyW * 0.5f + colW * 0.5f;
    for (int k = 0; k < leftToRight.Count; k++)
    {
        var chars = leftToRight[k].ToCharArray();
        var sb = new System.Text.StringBuilder(chars.Length * 2);
        for (int c = 0; c < chars.Length; c++)
        {
            if (c > 0) sb.Append('\n');
            sb.Append(chars[c]);
        }
        var t = MakeText(_detailOverlay, sb.ToString(), 34,
            new Vector2(x0 + k * colW, 90f), TextAnchor.MiddleCenter, colW, 600f);
        t.gameObject.name = "VertColumn"; // per-open columns, cleared on close
        t.color = SilhouetteBlack;
    }
    _detailOverlay.gameObject.SetActive(true);
    SetToast(" " + card.title + " ");
}
void OnDetailTapped()
{
    // Clear the per-open vertical columns, then hide the overlay.
    // EditMode gates walk this same path, so the immediate-destroy
    // branch keeps batch runs from stacking deferred duplicates.
    for (int i = _detailOverlay.childCount - 1; i >= 0; i--)
    {
        var child = _detailOverlay.GetChild(i);
        if (child.name == "VertColumn")
        {
            if (Application.isPlaying) UnityEngine.Object.Destroy(child.gameObject);
            else UnityEngine.Object.DestroyImmediate(child.gameObject);
        }
    }
    _detailBodyRaw = "";
    _detailColumns = 0;
    _detailOverlay.gameObject.SetActive(false);
    SyncView();
}
```

```

}
// —— per-intent sync ——
/// &summary&Idempotent full-state refresh: card slot faces from the
/// book (unlocked warm + verbatim title/category, locked dark),
/// the in-book counter, and the single archive_extra ad seat
/// (live only while the ad card is still locked - A2/A3).&/summary&
public void SyncView()
{
    if (_book == null) return;
    var cards = _book.Cards;
    for (int i = 0; i &_cardFills.Length; i++)
    {
        bool shown = i & cards.Count;
        SetActiveIfChanged(_cardFills[i].gameObject, shown);
        if (!shown) continue;
        string id = cards[i] != null ? cards[i].cardId : "";
        _cardIds[i] = id;
        bool open = _book.IsUnlocked(id);
        _cardFills[i].color = open ? WarmYellow : SlotDark;
        // Deep layout: title rides line one, the verbatim category
        // line two, led by a glyph chip whose mark IS the
        // category's first char (content-derived, zero hardcoded
        // case text in the view).
        _cardTexts[i].text = open ? cards[i].title : LockedLabel;
        _cardTexts[i].color = open ? SilhouetteBlack : PaperWhite;
        _cardTexts[i].alignment = open ? TextAnchor.MiddleLeft : TextAnchor.MiddleCenter;
        string cat = open ? (cards[i] != null ? cards[i].category : "") : "";
        _cardCats[i].text = cat;
        bool chipOn = open && cat.Length & 0;
        if (_cardChips[i] != null)
        {
            SetActiveIfChanged(_cardChips[i].gameObject, chipOn);
            if (chipOn)
            {
                var glyph = _cardChips[i].GetComponentInChildren&Text&();
                if (glyph != null) glyph.text = cat.Substring(0, 1);
            }
        }
        // T021: the seal stamp marks the unlocked face (layout
        // slice 1 - paper chip + seal; full card-face art later).
        if (_cardSeals[i] != null)
            SetActiveIfChanged(_cardSeals[i].gameObject, open);
    }
    if (_counterText != null)
        _counterText.text = "    " + _book.UnlockedCount + "/" + _book.TotalCount;
    // Ad seat: only the chapter's single archive_extra card rides
    // it, and only while that card is still locked.
    bool adLive = false;
    foreach (var c in cards)
        if (c != null && c.unlockType == "ad" && !_book.IsUnlocked(c.cardId)) { adLive = true; b

```

```

        SetActiveIfChanged(_adButton.gameObject, adLive);
    }
    // —— static face (built once) ——
    void BuildStaticFace()
    {
        var bg = MakeCanvasChild("Bg", SilhouetteBlack);
        bg.anchorMin = Vector2.zero;
        bg.anchorMax = Vector2.one;
        bg.anchoredPosition = Vector2.zero;
        bg.sizeDelta = Vector2.zero;
        bg.GetComponent<Image>().raycastTarget = false;
        // T031: nine-slice screen frame from the art catalog (X027 law) -
        // above the flat Bg base, under every interactive sibling; when
        // the catalog entry is absent the procedural Bg face stays.
        var frameArt = Sy.Core.ArtRuntime.Get("P08_ui_s02_chapter_frame");
        if (frameArt != null)
        {
            var frame = MakeCanvasChild("UiFrame", Color.white);
            frame.anchorMin = Vector2.zero;
            frame.anchorMax = Vector2.one;
            frame.anchoredPosition = Vector2.zero;
            frame.sizeDelta = Vector2.zero;
            var frameImg = frame.GetComponent<Image>();
            frameImg.sprite = frameArt;
            frameImg.type = Image.Type.Sliced;
            frameImg.raycastTarget = false;
        }
        _titleText = MakeText(_canvas.transform, "      ", 44, new Vector2(0f, 900f), TextAnchor.MiddleCenter, 1
        _titleText.color = WarmYellow;
        _moralityText = MakeText(_canvas.transform, "      0", 34, new Vector2(0f, 820f), TextAnchor.MiddleCenter, 60
        _moralityText.color = PaperWhite;
        // Chapter-select row: shipped chapter live, future chapter
        // locked behind its free-pass window (launch face).
        _chapterOneButton = MakeBar("ChapterOne", "      ",
            new Vector2(0f, 690f), new Vector2(980f, 120f), NetherBlue, () => ChapterTapped?.Invoke(_chapterOneId
        _chapterTwoButton = MakeBar("ChapterTwo", "      ",
            new Vector2(0f, 550f), new Vector2(980f, 120f), ChapterLocked, () => { }); // locked: dead face, no
        _chapterTwoButton.interactable = false;
        _freePassText = MakeText(_canvas.transform, "      ", 26, new Vector2(0f, 460f), TextAnchor.MidleC
        _freePassText.color = TalismanGold;
        // Case-code share seat (the X372 promise: the settled-case
        // code lands on this page; host-fed, deterministic).
        _codeText = MakeText(_canvas.transform, "", 40, new Vector2(0f, 380f), TextAnchor.MiddleCenter, 1000f, 60f);
        _codeText.color = TalismanGold;
        MakeText(_canvas.transform, "      ", 36, new Vector2(-420f, 270f), TextAnchor.MiddleCenter, 160f, 50f).color
        _counterText = MakeText(_canvas.transform, "      0/5", 30, new Vector2(370f, 270f), TextAnchor.MiddleCenter,
        _counterText.color = PaperWhite;
        for (int i = 0; i < _cardFills.Length; i++)
        {
            int idx = i;

```

```

        BuildSettlePanel();
        SetHint(" .....");
    }
    /// <summary>
    /// Deterministic presentation clock (T008 slice 2b seam): advances every
    /// beat-relevant animation by dt. Play mode: Update() feeds Time.deltaTime
    /// (same cadence the old coroutines had). Editor gates (Redline11RtCheck):
    /// fed directly, giving exact control over mid-course beat moments.
    /// Idempotent for dt=0 (asserts stay deterministic without frame time).
    /// </summary>
    public void PresentTick(float dt)
    {
        _presentTime += dt;
        // T019: paper-figure tilt beat through the shared micro-motion
        // template (law unchanged: still figure until the arm window).
        _figureMotion.Tick(dt);
        _figureMotion.Apply(_paperFigure);
        // Candle breathing (0.5Hz) + realm temperature color flip.
        float breathe = 0.5f + 0.5f * Mathf.Sin(_presentTime * Mathf.PI);
        _candleGlow.localScale = Vector3.one * (0.96f + 0.08f * breathe);
        _candleImage.color = _realmCold ? CandleCold : CandleWarm;
        // Drum shake: first kick lands inside PlayDrumBeat itself, ticks decay it.
        if (_shakeActive)
        {
            _shakeT += dt;
            var rt = (RectTransform)_canvas.transform;
            if (_shakeT >= ShakeSeconds)
            {
                _shakeActive = false;
                rt.anchoredPosition = _shakeOrigin;
            }
            else
            {
                rt.anchoredPosition = _shakeOrigin + (Vector2)(UnityEngine.Random.insideUnitCircle * 10f);
            }
        }
        // Realm-switch overlay: warm->cold lerp across the pinned switch face.
        if (_switchActive)
        {
            _switchT += dt;
            if (_switchT >= _sim.Tuning.SwitchSeconds)
            {
                _switchActive = false;
                _switchOverlay.gameObject.SetActive(false);
            }
            else
            {
                float k = Mathf.Clamp01(_switchT / _sim.Tuning.SwitchSeconds);
                _switchImage.color = Color.Lerp(SwitchWarm, SwitchCold, k);
            }
        }
    }

```

```
    }
}
void Update()
{
    PresentTick(Time.deltaTime);
}
// -----
void BuildScene()
{
    var bg = MakeCanvasChild("Bg", SilhouetteBlack);
    bg.anchorMin = Vector2.zero;
    bg.anchorMax = Vector2.one;
    bg.anchoredPosition = Vector2.zero;
    bg.sizeDelta = Vector2.zero;
    // X027 generated art: swap the solid Bg for the shadow-play stage painting.
    var bgImg = bg.GetComponent<Image>();
    var bgArt = Sy.Core.ArtRuntime.Get("P08_scene_01");
    if (bgArt != null)
    {
        bgImg.sprite = bgArt;
        bgImg.color = Color.white;
        bgImg.preserveAspect = true;
        bgImg.raycastTarget = false;
    }
    //
    var curtain = MakeCanvasChild("Curtain", CurtainRed);
    curtain.anchoredPosition = new Vector2(0, 820);
    curtain.sizeDelta = new Vector2(1080, 280);
    //      —————→
    _candleGlow = MakeCanvasChild("CandleGlow", CandleWarm);
    _candleImage = _candleGlow.GetComponent<Image>();
    _candleGlow.anchoredPosition = new Vector2(0, -60);
    _candleGlow.sizeDelta = new Vector2(860, 860);
    //      M1
    var lanternGo = new GameObject("Lantern", typeof(RectTransform), typeof(Image), typeof(Button));
    lanternGo.transform.SetParent(_canvas.transform, false);
    _lanternRt = (RectTransform)lanternGo.transform;
    _lanternRt.anchorMin = _lanternRt.anchorMax = new Vector2(0.5f, 0.5f);
    _lanternRt.anchoredPosition = new Vector2(-320, 220);
    _lanternRt.sizeDelta = new Vector2(150, 190);
    var lanternImg = lanternGo.GetComponent<Image>();
    lanternImg.sprite = _square;
    lanternImg.color = WarmYellow;
    MakeText(_lanternRt, " ", 66, Vector2.zero, TextAnchor.MiddleCenter, 150, 190).color = SilhouetteBlack;
    lanternGo.GetComponent<Button>().onClick.AddListener(() => LanternTapped?.Invoke());
    //      M0      —X027 generated art when present
    _paperFigure = MakeCanvasChild("PaperFigure", PaperWhite);
    _paperFigure.anchoredPosition = new Vector2(340, 160);
    _paperFigure.sizeDelta = new Vector2(90, 150);
    var paperImg = _paperFigure.GetComponent<Image>();
```

```

var paperArt = Sy.Core.ArtRuntime.Get("P08_unit_01");
if (paperArt != null)
{
    _paperFigure.sizeDelta = new Vector2(170, 280);
    paperImg.sprite = paperArt;
    paperImg.color = Color.white;
    paperImg.preserveAspect = true;
}
else
{
    paperImg.sprite = _square;
    paperImg.color = PaperWhite;
}
MakeText(_paperFigure, " ", 48, Vector2.zero, TextAnchor.MiddleCenter, 90, 150).color = SilhouetteBlack;
//      M2.5      —"      "
var corpseGo = new GameObject("Corpse", typeof(RectTransform), typeof(Image), typeof(Button));
corpseGo.transform.SetParent(_canvas.transform, false);
_corpseRt = (RectTransform)corpseGo.transform;
_corpseRt.anchorMin = _corpseRt.anchorMax = new Vector2(0.5f, 0.5f);
_corpseRt.anchoredPosition = new Vector2(0, -60);
_corpseRt.sizeDelta = new Vector2(240, 400);
var corpseImg = corpseGo.GetComponent<Image>();
corpseImg.sprite = _square;
corpseImg.color = new Color(0.13f, 0.15f, 0.18f);
MakeText(_corpseRt, " \n      ", 36, Vector2.zero, TextAnchor.MiddleCenter, 240, 400).color = PaperWhite;
corpseGo.GetComponent<Button>().onClick.AddListener(() => CorpseTapped?.Invoke());
corpseGo.SetActive(false);
//      M4      —"      "
var ghostGo = new GameObject("Ghost", typeof(RectTransform), typeof(Image), typeof(Button));
ghostGo.transform.SetParent(_canvas.transform, false);
_ghostRt = (RectTransform)ghostGo.transform;
_ghostRt.anchorMin = _ghostRt.anchorMax = new Vector2(0.5f, 0.5f);
_ghostRt.anchoredPosition = new Vector2(0, -140);
_ghostRt.sizeDelta = new Vector2(240, 240);
var ghostImg = ghostGo.GetComponent<Image>();
ghostImg.sprite = _square;
ghostImg.color = NetherBlue;
MakeText(_ghostRt, " \n      ", 40, Vector2.zero, TextAnchor.MiddleCenter, 240, 240).color = PaperWhite;
ghostGo.GetComponent<Button>().onClick.AddListener(() => GhostTapped?.Invoke());
ghostGo.SetActive(false);
//      M3      —
var burnGo = new GameObject("BurnButton", typeof(RectTransform), typeof(Image), typeof(Button));
burnGo.transform.SetParent(_canvas.transform, false);
var burnRt = (RectTransform)burnGo.transform;
burnRt.anchorMin = burnRt.anchorMax = new Vector2(0.5f, 0f);
burnRt.anchoredPosition = new Vector2(0, 120);
burnRt.sizeDelta = new Vector2(460, 130);
var burnImg = burnGo.GetComponent<Image>();
burnImg.sprite = _square;
burnImg.color = CurtainRed;

```

```

        MakeText(burnRt, "    ", 52, Vector2.zero, TextAnchor.MiddleCenter, 460, 130).color = PaperWhite;
        _burnButton = burnGo.GetComponent<Button>();
        _burnButton.interactable = true;
        _burnButton.onClick.AddListener(() => BurnTapped?.Invoke());
        burnGo.SetActive(false);
    }
    void BuildHud()
    {
        var root = _canvas.transform;
        _hintText = MakeText(root, "", 42, new Vector2(0, -840), TextAnchor.MiddleCenter, 1020, 160);
        _hintText.color = PaperWhite;
        //          /
        MakeText(root, "    ", 36, new Vector2(-440, -700), TextAnchor.MiddleCenter, 160, 50).color = WarmYellow;
        for (int i = 0; i < _clueSlots.Length; i++)
        {
            var slot = MakeCanvasChild("ClueSlot_" + i, new Color(0.12f, 0.10f, 0.08f, 0.9f));
            slot.anchoredPosition = new Vector2(-440, -700 - i * 130);
            slot.sizeDelta = new Vector2(110, 110);
            _clueSlots[i] = slot.GetComponent<Image>();
        }
    }
    void BuildDrumPanel()
    {
        _drumPanel = MakeCanvasChild("DrumPanel", new Color(0, 0, 0, 0));
        _drumPanel.GetComponent<Image>().color = new Color(0, 0, 0, 0);
        _drumCountText = MakeText(_drumPanel, "", 80, new Vector2(0, 420), TextAnchor.MiddleCenter, 800, 120);
        _drumCountText.color = WarmYellow;
        for (int i = 0; i < 10; i++)
        {
            int digit = i;
            var go = new GameObject("Digit_" + i, typeof(RectTransform), typeof(Image), typeof(Button));
            go.transform.SetParent(_canvas.transform, false);
            var rt = (RectTransform)go.transform;
            rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0f);
            rt.anchoredPosition = new Vector2(-405 + i * 90, 300);
            rt.sizeDelta = new Vector2(80, 90);
            var img = go.GetComponent<Image>();
            img.sprite = _square;
            img.color = CurtainRed;
            _digitImages[i] = img;
            MakeText(go.transform, i.ToString(), 44, Vector2.zero, TextAnchor.MiddleCenter, 80, 90).color = PaperWhite;
            _digitButtons[i] = go.GetComponent<Button>();
            _digitButtons[i].onClick.AddListener(() => DigitTapped?.Invoke(digit));
        }
    }
    void BuildVerdictPanel()
    {
        _verdictPanel = MakeCanvasChild("VerdictPanel", new Color(0, 0, 0, 0));
        _verdictPanel.GetComponent<Image>().color = new Color(0, 0, 0, 0);
        MakeText(_verdictPanel, "    ", 64, new Vector2(0, 620), TextAnchor.MiddleCenter, 1000, 100).color = P

```

```

for (int i = 0; i < NewbieConfig.VerdictOptions.Length; i++)
{
    int index = i;
    var go = new GameObject("Verdict_" + i, typeof(RectTransform), typeof(Image), typeof(Button));
    go.transform.SetParent(_verdictPanel, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = new Vector2(-250 + i * 500, 480);
    rt.sizeDelta = new Vector2(420, 150);
    var img = go.GetComponent<Image>();
    img.sprite = _square;
    img.color = i == NewbieConfig.VerdictCorrect ? NetherBlue : new Color(0.3f, 0.12f, 0.10f);
    MakeText(go.transform, NewbieConfig.VerdictOptions[i], 48, Vector2.zero, TextAnchor.MiddleCenter, 420, 1
    go.GetComponent<Button>().onClick.AddListener(() => VerdictTapped?.Invoke(index));
}
_verdictPanel.gameObject.SetActive(false);
}

void BuildSwitchOverlay()
{
    _switchOverlay = MakeCanvasChild("SwitchOverlay", new Color(0, 0, 0, 0));
    _switchImage = _switchOverlay.GetComponent<Image>();
    _switchOverlay.anchorMin = Vector2.zero;
    _switchOverlay.anchorMax = Vector2.one;
    _switchOverlay.anchoredPosition = Vector2.zero;
    _switchOverlay.sizeDelta = Vector2.zero;
    MakeText(_switchOverlay, " · ", 90, new Vector2(0, 0), TextAnchor.MiddleCenter, 1000, 140).color = Paper
    _switchOverlay.gameObject.SetActive(false);
}

void BuildSettlePanel()
{
    _settlePanel = MakeOverlayPanel("SettlePanel", new Color(0.03f, 0.03f, 0.04f, 0.96f));
    MakeText(_settlePanel, " ", 96, new Vector2(0, 560), TextAnchor.MiddleCenter, 1040, 120).color = WarmYell
    MakeText(_settlePanel,
        " · " + _sim.Tuning.YindeGain + "\n" + NewbieConfig.ArchiveEntry +
        "\n<size=42> — </size>",
        44, new Vector2(0, 340), TextAnchor.MiddleCenter, 1020, 380).color = PaperWhite;
    MakeActionButton(_settlePanel, " ", new Vector2(-260, -300), () => UnityEngine.SceneManagement.SceneMa
    MakeActionButton(_settlePanel, " ", new Vector2(260, -300), () => UnityEngine.SceneManagement.SceneMa
    _settlePanel.gameObject.SetActive(false);
}

// ————
public void SyncView(CaseSim sim)
{
    var phase = sim.Phase;
    SetActiveIfChanged(_lanternRt.gameObject, phase == CaseSim.CasePhase.AwaitLantern);
    SetActiveIfChanged(_corpseRt.gameObject, phase == CaseSim.CasePhase.AwaitCorpse);
    SetActiveIfChanged(_ghostRt.gameObject, phase == CaseSim.CasePhase.AwaitClues);
    SetActiveIfChanged(_burnButton.gameObject, phase == CaseSim.CasePhase.AwaitBurn);
    bool inputOk = phase == CaseSim.CasePhase.AwaitInput;
    for (int i = 0; i < 10; i++)

```

```

        SetActiveIfChanged(_digitButtons[i].gameObject, inputOk);
        SetActiveIfChanged(_verdictPanel.gameObject, phase == CaseSim.CasePhase.AwaitVerdict);
    }
    static void SetActiveIfChanged(GameObject go, bool active)
    {
        if (go.activeSelf != active) go.SetActive(active);
    }
    // —— ——
    /// <summary> + + M2 — </summary>
    public void PlayDrumBeat(int index)
    {
        _drumCountText.text = " x" + (index + 1);
        PlayTone(NewbieConfig.DrumToneFreq, NewbieConfig.DrumToneSeconds);
        // Shake kick lands on the beat itself (same instant the old
        // coroutine's first synchronous segment displaced the canvas);
        // PresentTick decays it for ShakeSeconds afterwards.
        _shakeActive = true;
        _shakeT = 0f;
        var rt = (RectTransform)_canvas.transform;
        _shakeOrigin = rt.anchoredPosition;
        rt.anchoredPosition = _shakeOrigin + (Vector2)(UnityEngine.Random.insideUnitCircle * 10f);
    }
    /// <summary> </summary>
    public void PlayBuzz()
    {
        PlayTone(NewbieConfig.BuzzToneFreq, NewbieConfig.BuzzToneSeconds);
    }
    /// <summary> </summary>
    public void PlayLockOpen()
    {
        PlayTone(NewbieConfig.ChimeToneFreq, NewbieConfig.ChimeToneSeconds * 0.5f);
    }
    /// <summary>M3 0.8s → + </summary>
    public void PlayRealmSwitch()
    {
        _realmCold = true;
        // Warm phase renders immediately (reversal START face); the
        // warm->cold trajectory advances in PresentTick only.
        _switchActive = true;
        _switchT = 0f;
        _switchOverlay.gameObject.SetActive(true);
        _switchImage.color = SwitchWarm;
    }
    /// <summary> 0 </summary>
    public void OnCorpseInspected()
    {
        _clueSlots[0].color = WarmYellow;
        PlayTone(NewbieConfig.ChimeToneFreq, NewbieConfig.ChimeToneSeconds * 0.5f);
    }
    /// <summary> 1 </summary>

```

```

public void OnGhostSpoken()
{
    _clueSlots[1].color = NetherBlue;
    PlayTone(NewbieConfig.ChimeToneFreq, NewbieConfig.ChimeToneSeconds);
}
/// <summary>          </summary>
public void PlayVerdictChime()
{
    PlayTone(NewbieConfig.ChimeToneFreq, NewbieConfig.ChimeToneSeconds);
}
// ----- HUD -----
public void SetHint(string hint) => _hintText.text = hint;
public void ShowSettle()
{
    _settlePanel.gameObject.SetActive(true);
}
// ----- + $ -----
void PlayTone(float freq, float seconds)
{
    if (_audio == null) _audio = gameObject.AddComponent<AudioSource>();
    _audio.PlayOneShot(MakeTone(freq, seconds));
}
static AudioClip MakeTone(float freq, float seconds)
{
    int sr = NewbieConfig.ToneSampleRate;
    int n = Mathf.Max(1, (int)(sr * seconds));
    var data = new float[n];
    for (int i = 0; i < n; i++)
    {
        float k = 1f - (float)i / n;
        data[i] = Mathf.Sin(2f * Mathf.PI * freq * i / sr) * k;
    }
    var clip = AudioClip.Create("tone", n, 1, sr, false);
    clip.SetData(data, 0);
    return clip;
}
// -----
RectTransform MakeCanvasChild(string name, Color color)
{
    var go = new GameObject(name, typeof(RectTransform), typeof(Image));
    go.transform.SetParent(_canvas.transform, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    var img = rt.GetComponent<Image>();
    img.sprite = _square;
    img.color = color;
    return rt;
}
RectTransform MakeOverlayPanel(string name, Color color)
{

```

```
var go = new GameObject(name, typeof(RectTransform), typeof(Image));
go.transform.SetParent(_canvas.transform, false);
var rt = (RectTransform)go.transform;
rt.anchorMin = Vector2.zero;
rt.anchorMax = Vector2.one;
rt.offsetMin = rt.offsetMax = Vector2.zero;
var image = go.GetComponent<Image>();
image.sprite = _square;
image.color = color;
return rt;
}

void MakeActionButton(RectTransform parent, string text, Vector2 pos, Action onClick)
{
    var go = new GameObject(text, typeof(RectTransform), typeof(Image), typeof(Button));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = new Vector2(420, 120);
    var image = go.GetComponent<Image>();
    image.sprite = _square;
    image.color = CurtainRed;
    MakeText(go.transform, text, 46, Vector2.zero, TextAnchor.MiddleCenter, 420, 120).color = PaperWhite;
    go.GetComponent<Button>().onClick.AddListener(() => onClick());
}

static Text MakeText(Transform parent, string text, int size, Vector2 pos, TextAnchor anchor, float w, float h)
{
    var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = new Vector2(w, h);
    var t = go.GetComponent<Text>();
    t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
    t.text = text;
    t.fontSize = size;
    t.alignment = anchor;
    t.color = new Color(0.91f, 0.89f, 0.85f);
    t.supportRichText = true;
    return t;
}

static Sprite MakeSquareSprite()
{
    var tex = new Texture2D(8, 8, TextureFormat.RGBA32, false);
    var px = tex.GetPixels32();
    for (int i = 0; i < px.Length; i++) px[i] = new Color32(255, 255, 255, 255);
    tex.SetPixels32(px);
    tex.Apply(false, false);
    return Sprite.Create(tex, new Rect(0, 0, 8, 8), new Vector2(0.5f, 0.5f), 8f);
}
```

```

    }
}
}
// NewbieConfig.cs
namespace Sy.P08.Newbie
{
    /// <summary>
    /// P08 . V0.2 — X009+IConfigService Design/configs/P08 JSON
    /// / —" "/ 2 1 / /
    /// 06 08 M0-M5
    /// #8B0000 x #FFB347 x #0D0D0D x #F5E6C8 x #1B3A4B L≥25
    /// </summary>
    public static class NewbieConfig
    {
        public const string GameTitle = " ";
        public const string GameSlogan = " ";
        // SW5 flip X366: CaseSweepCheck 5 x8seed = e P50 9.18s· 1.75s· PASS
        public const float IntroSeconds = 2.6f; // M0 +
        public const float DrumBeatInterval = 0.65f; // M2 - -
        public const float DrumFirstDelay = 0.4f;
        public const int DrumCount = 3; // = + =
        public const int LockAnswer = 3;
        public const float SwitchSeconds = 0.55f; // M3 → 0.5s
        // — §4.2 ③ —
        public static readonly string[] ClueNames = { " ", " ", " " };
        public const string CorpseInspectText = " — ";
        public const string GhostSpeakText = " — ..... — ";
        // — §4.2 ② 1 2 1 → = —
        public const int VerdictCorrect = 0; // A
        public const string VerdictCorrectText = " + — ";
        public const string VerdictWrongGuidance = "B — ";
        public static readonly string[] VerdictOptions = { "A ", "B " };
        // — §4.3 = + 20 —
        public const int YindeGain = 10; // +10
        public const string ArchiveEntry = " #1 — ";
        public const float MaxTutorialSeconds = 210f; // ≤3.5min 06 ②
        // — / / — § —
        public const int ToneSampleRate = 44100;
        public const float DrumToneFreq = 92f;
        public const float DrumToneSeconds = 0.28f;
        public const float BuzzToneFreq = 220f;
        public const float BuzzToneSeconds = 0.18f;
        public const float ChimeToneFreq = 660f;
        public const float ChimeToneSeconds = 0.4f;
    }
}
// NewbieTuning.cs
using System.Collections.Generic;
using Sy.Core;
using UnityEngine;

```

```

namespace Sy.P08.Newbie
{
    /// <summary>
    /// T008 SW-1      8      readonly struct=
    ///      Design/configs/P08/cfg_newbie_tuning.json    v1.0.0    Resources
    ///      BuiltIn()    NewbieConfig    =    build JSON    ·X032
    ///      X363    § v1.0    /b (firstDelay0.4+interval0.65)/c (intro2.6)/d (switch0.55)/e bcd
    /// </summary>
    public readonly struct NewbieTuning
    {
        public readonly int Version;
        public readonly string Label;          // ConfigPin    "v1.0"/"b"/"c"/"d"/"e"
        public readonly float IntroSeconds;     // M0            ≥2.6s=
        public readonly float DrumFirstDelay;    // M2            ≥0.3s
        public readonly float DrumBeatInterval; // M2
        public readonly int DrumCount;          //
        public readonly int LockAnswer;         // =
        public readonly float SwitchSeconds;    // M3
        public readonly int YindeGain;          // M5
        public readonly float MaxTutorialSeconds; //      06      @≤210s
        public NewbieTuning(int version, string label, float introSeconds, float drumFirstDelay,
            float drumBeatInterval, int drumCount, int lockAnswer, float switchSeconds,
            int yindeGain, float maxTutorialSeconds)
        {
            Version = version;
            Label = label;
            IntroSeconds = introSeconds;
            DrumFirstDelay = drumFirstDelay;
            DrumBeatInterval = drumBeatInterval;
            DrumCount = drumCount;
            LockAnswer = lockAnswer;
            SwitchSeconds = switchSeconds;
            YindeGain = yindeGain;
            MaxTutorialSeconds = maxTutorialSeconds;
        }
    }
    /// <summary>
    ///      Live()=      ·      /Candidate(arm)=      /
    /// SetActiveForTests()=Editor      SW-2 CaseSweepCheck      ·
    /// shipped    Candidate("v1.0")    —      JSON    SW4 shipped==winner
    /// </summary>
    public static class NewbieTuningTable
    {
        //      SW5    =      NewbieConfig
        public static NewbieTuning BuiltIn() => new NewbieTuning(
            1, "v1.0",
            NewbieConfig.IntroSeconds, NewbieConfig.DrumFirstDelay, NewbieConfig.DrumBeatInterval,
            NewbieConfig.DrumCount, NewbieConfig.LockAnswer, NewbieConfig.SwitchSeconds,
            NewbieConfig.YindeGain, NewbieConfig.MaxTutorialSeconds);
        static NewbieTuning? _testOverride;
    }
}

```

```

/// <summary> build IO JSON= </summary>
public static NewbieTuning Live() => _testOverride ?? BuiltIn();
/// <summary>Editor batch null </summary>
public static void SetActiveForTests(NewbieTuning? tuning) => _testOverride = tuning;
/// <summary> Resources "v1.0" / null </summary>
public static NewbieTuning? Candidate(string profile)
{
    if (profile == "v1.0") return BuiltIn();
    var asset = Resources.Load<TextAsset>("cfg_newbie_tuning");
    if (asset == null) return null;
    var profiles = ArchiveFromJson(asset.text);
    if (profiles == null || !profiles.TryGetValue(profile, out var t)) return null;
    return t;
}
/// <summary> profiles / —SW-2
/// null 0 loud fail </summary>
public static Dictionary<string, NewbieTuning> ArchiveFromJson(string json)
{
    if (!(MiniJson.Deserialize(json) is Dictionary<string, object> root)) return null;
    if (!root.TryGetValue("profiles", out var node) || !(node is Dictionary<string, object> profiles)) return null;
    var table = new Dictionary<string, NewbieTuning>();
    foreach (var kv in profiles)
    {
        if (!(kv.Value is Dictionary<string, object> p)) continue;
        string label = kv.Key;
        if (p.TryGetValue("label", out var l) && l is string s && !string.IsNullOrEmpty(s)) label = s;
        table[kv.Key] = new NewbieTuning(
            AsInt(p, "version"),
            label,
            AsFloat(p, "introSeconds"),
            AsFloat(p, "drumFirstDelay"),
            AsFloat(p, "drumBeatInterval"),
            AsInt(p, "drumCount"),
            AsInt(p, "lockAnswer"),
            AsFloat(p, "switchSeconds"),
            AsInt(p, "yindeGain"),
            AsFloat(p, "maxTutorialSeconds"));
    }
    return table;
}
static int AsInt(Dictionary<string, object> node, string key)
{
    if (!node.TryGetValue(key, out var v)) return 0;
    return v switch
    {
        long l => (int)l,
        double d => (int)d,
        _ => 0,
    };
}

```

```

static float AsFloat(Dictionary<string, object>; node, string key)
{
    if (!node.TryGetValue(key, out var v)) return 0f;
    return v switch
    {
        long l => 1,
        double d => (float)d,
        _ => 0f,
    };
}

}

// PuppetMicroMotion.cs
using UnityEngine;
namespace Sy.P08.Newbie
{
    /// <summary>
    /// T019 micro-motion template: one shared deterministic motion profile for the
    /// four paper-shadow jointed puppets (unit_01 paper figure + yincha / female /
    /// judge). Three channels - the uncanny head-tilt beat (always-on law for the
    /// M0 figure), an optional breath bob and an optional joint sway - all driven
    /// by the caller's deterministic presentation clock (PresentTick seam): zero
    /// sim state, zero Time.deltaTime reads, idempotent for dt = 0.
    /// The M0 paper figure runs the tilt channel ALONE: the beat's power is the
    /// perfectly still figure that suddenly tilts (uncanny-valley law, 06 script
    /// s0: first tilt <= T+3.0s, Redline11RtCheck R6). Breath and sway arm the
    /// other three puppets on their story surfaces with per-puppet phase offsets
    /// so the group never moves in lockstep; keep phase within [0, 0.8]s so every
    /// puppet's first tilt still lands inside the 3.0s s0 beat window.
    /// Sequence-frame channels (blink / limb frames) attach when the multi-joint
    /// layered art lands (art doc 7.3 pending refinement); this template owns the
    /// transform law so frame variants only swap sprites, never re-time beats.
    /// </summary>
    public sealed class PuppetMicroMotion
    {
        public const float TiltCycleSeconds = 2.2f; // one tilt per cycle (validated CaseView preview)
        public const float TiltArmSeconds = 2.0f; // still figure until the arm window opens
        public const float TiltAngle = 0.5f; // uncanny beat: a hair of tilt
        const float BreathHz = 0.5f; // matches the candle glow (stage unity)
        const float BreathBobPixels = 3f;
        const float SwayHz = 0.25f;
        const float SwayAngle = 0.3f;
        readonly float _phase;
        float _wobble; // tilt cycle clock, original accumulation law
        bool _hasHome;
        Vector2 _home;
        public bool Breath { get; }
        public bool Sway { get; }
        public bool Tilt { get; }
        /// <param name="phase">Per-puppet offset, clamped to [0, 0.8]s so the

```

```

    /// first tilt of every puppet in the group stays inside the 3.0s s0 window.</param>
    public PuppetMicroMotion(bool breath, bool sway, bool tilt, float phase = 0f)
    {
        Breath = breath;
        Sway = sway;
        Tilt = tilt;
        _phase = Mathf.Clamp(phase, 0f, 0.8f);
    }
    /// <summary>Advance the motion clocks by dt (caller = the deterministic
    /// presentation seam; idempotent for dt = 0).</summary>
    public void Tick(float dt)
    {
        _wobble += dt;
        if (_wobble > TiltCycleSeconds) _wobble = 0f;
    }
    /// <summary>Apply the enabled channels to the puppet root. Rotation is
    /// exactly identity before the tilt arm window - the beat's stillness law
    /// the RT gate reads (R6: localRotation != identity marks the beat).</summary>
    public void Apply(RectTransform puppet)
    {
        float t = _phase + _wobble;
        float tilt = Tilt && _wobble > TiltArmSeconds ? TiltAngle : 0f;
        float sway = Sway ? Mathf.Sin(t * ((2f * Mathf.PI) * SwayHz)) * SwayAngle : 0f;
        float angle = tilt + sway;
        puppet.localRotation = angle == 0f
            ? Quaternion.identity
            : Quaternion.Euler(0f, 0f, angle);
        if (Breath)
        {
            if (!_hasHome)
            {
                _home = puppet.anchoredPosition;
                _hasHome = true;
            }
            float bob = Mathf.Sin(t * ((2f * Mathf.PI) * BreathHz)) * BreathBobPixels;
            puppet.anchoredPosition = _home + new Vector2(0f, bob);
        }
    }
}
// TutorialFlow.cs
using System;
namespace Sy.P08.Newbie
{
    /// <summary>
    ///     evt_newbie_step 1-5 ① ② ③ ④ ⑤ +
    ///     CaseSim / / / / / / / / /
    /// Play headless NewbieFlowCheck
    /// </summary>
    public sealed class TutorialFlow

```

```
{
    public int Step { get; private set; }
    readonly Action<string> _hint;
    readonly Action<int> _onStep;
    readonly CaseSim _sim;    // T008 SW-1
    public TutorialFlow(CaseSim sim, Action<string> hint, Action<int> onStep)
    {
        _sim = sim;
        _hint = hint ?? delegate { };
        _onStep = onStep ?? delegate { };
        sim.PhaseChanged += OnPhaseChanged;
        sim.LanternTaken += OnLanternTaken;
        sim.DrumBeat += OnDrumBeat;
        sim.WrongInput += OnWrongInput;
        sim.LockOpenedDone += OnLockOpened;
        sim.CorpseInspected += OnCorpseInspected;
        sim.RealmSwitchedDone += OnRealmSwitched;
        sim.GhostSpoken += OnGhostSpoken;
        sim.CluesComplete += OnCluesComplete;
        sim.VerdictGuidance += OnVerdictGuidance;
        sim.VerdictCorrect += OnVerdictCorrect;
        _hint("    —    ");
    }
    void Advance(int step)
    {
        if (Step >= step) return;
        Step = step;
        _onStep(step);
    }
    void OnLanternTaken()
    {
        Advance(1);
        _hint("    —    ");
    }
    void OnDrumBeat(int index)
    {
        _hint("    " + (index + 1) + " — ");
    }
    void OnWrongInput(int digit)
    {
        _hint("    —    " + digit + " ");
    }
    void OnLockOpened()
    {
        Advance(2);
        _hint("    —    ");
    }
    void OnCorpseInspected()
    {
        _hint(NewbieConfig.CorpseInspectText);
    }
}
```

```
    }
    void OnRealmSwitched()
    {
        Advance(3);
        _hint("    —    ");
    }
    void OnGhostSpoken()
    {
        _hint(NewbieConfig.GhostSpeakText);
    }
    void OnCluesComplete()
    {
        Advance(4);
        _hint("    —    ");
    }
    void OnVerdictGuidance(int option)
    {
        _hint(NewbieConfig.VerdictWrongGuidance);
    }
    void OnVerdictCorrect()
    {
        Advance(5);
        _hint(NewbieConfig.VerdictCorrectText + "  +" + _sim.Tuning.YindeGain + "  ·  " + NewbieConfig.ArchiveEntry);
    }
    void OnPhaseChanged(CaseSim.CasePhase phase)
    {
        switch (phase)
        {
            case CaseSim.CasePhase.AwaitLantern:
                _hint("    —    ");
                break;
            case CaseSim.CasePhase.AwaitInput:
                _hint("    ");
                break;
            case CaseSim.CasePhase.AwaitCorpse:
                _hint("    —    ");
                break;
            case CaseSim.CasePhase.AwaitBurn:
                _hint("    —    ");
                break;
            case CaseSim.CasePhase.Switching:
                _hint("    —    ");
                break;
            case CaseSim.CasePhase.AwaitVerdict:
                _hint("    —    ");
                break;
        }
    }
}
}
```

```
// UIDemoView.cs
using System;
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.UI;
using UnityEngine.EventSystems;
using Sy.Core;
namespace Sy.P08
{
    /// <summary>
    /// U180 UIDemo: product-grade UGUI assembly proof (7-layer stack), fully
    /// procedural on top of the X027 two-state art law. Two mount paths:
    /// (a) scene Play - Awake builds its own camera + world-space canvas +
    /// EventSystem; (b) headless gate - MountOn(canvas) builds onto an
    /// existing rig canvas. All copy comes from Resources/cfg_uidemo.json
    /// (zero hardcoded text; X364 double face with Design/configs/P08).
    /// Layer stack - sibling order == render order (asserted by gate):
    /// L0 Bg / L1 Hud / L2 Panel / L3 Popup / L4 Toast / L5 Guide / L6 Top
    /// Raycast policy is part of the product-grade contract:
    /// Bg dead / Hud live-buttons-only / Panel live / Popup mask live
    /// Toast dead / Guide dead / Top dead.
    /// </summary>
    public sealed class UIDemoView : MonoBehaviour
    {
        // House palette tokens (AboutPanel / X373 canary anchors).
        static readonly Color CurtainRed = new Color(0.545f, 0f, 0f);
        static readonly Color WarmYellow = new Color(1f, 0.702f, 0.278f);
        static readonly Color PaperWhite = new Color(0.961f, 0.902f, 0.784f);
        static readonly Color Ink = new Color(0.13f, 0.09f, 0.10f);
        [Serializable]
        public sealed class CardEntry { public string name; public string icon; }
        [Serializable]
        public sealed class UIDemoConfig
        {
            public string screenTitle;
            public string hudChipLabel;
            public string hudHint;
            public string panelHeader;
            public string primaryLabel;
            public string secondaryLabel;
            public string popupTitle;
            public string popupBody;
            public string popupConfirm;
            public string popupCancel;
            public string toastClue;
            public string toastConfirm;
            public string toastSecondary;
            public string guideText;
            public string guideToggle;
        }
    }
}
```

```

        public string topChip;
        public List<CardEntry> cards;
    }
    public Canvas target; // gate-injected canvas; null => self-built rig
    public readonly List<string> layerOrder = new List<string>();
    UIDemoConfig _cfg;
    Sprite _square, _curtain;
    Canvas _canvas;
    readonly List<RectTransform> _layers = new List<RectTransform>();
    // Gate-facing faces.
    public Image PanelFrame;
    public Button PrimaryButton, SecondaryButton;
    public List<Button> CardButtons = new List<Button>();
    public Image PopupMask;
    public GameObject PopupRoot, PopupDialog, GuideRoot;
    public Image ToastStrip;
    public Text ToastText, TopChipText;
    static readonly string[] LayerNames =
        { "L0_Bg", "L1_Hud", "L2_Panel", "L3_Popup", "L4_Toast", "L5_Guide", "L6_Top" };
    const string FrameStem = "P08_ui_s02_chapter_frame";
    // Editor-time direct-path fallback dirs (probed 2026-09-24: the
    // ArtCatalog ScriptableObject loads NULL headless - typed AND untyped
    // - while the sprites themselves load fine; the house null-safe law
    // was masking this in pixel-canary gates. Production runtime keeps
    // the catalog/Resources channel; this fallback serves editor and
    // batch contexts until the lane fixes the catalog headless load.)
    static readonly string[] EditorArtDirs = { "Share", "Icons" };
    static Sprite ResolveArt(string stem)
    {
        var s = ArtRuntime.Get(stem);
        if (s != null) return s;
    }
    #if UNITY_EDITOR
        if (!string.IsNullOrEmpty(stem))
        {
            for (int i = 0; i < EditorArtDirs.Length; i++)
            {
                string path = "Assets/_Game/Art/" + EditorArtDirs[i] + "/" + stem + ".png";
                var byPath = UnityEditor.AssetDatabase.LoadAssetAtPath<Sprite>(path);
                if (byPath == null)
                {
                    // Some imports keep the sprite as a sub-asset next to a
                    // Texture2D main asset (probed 2026-09-24): sweep subs.
                    var subs = UnityEditor.AssetDatabase.LoadAllAssetsAtPath(path);
                    if (subs != null)
                        for (int j = 0; j < subs.Length; j++)
                            if (subs[j] is Sprite sp) { byPath = sp; break; }
                }
                if (byPath != null) return byPath;
            }
        }
    }

```

```
#endif
    return null;
}
void Awake()
{
    // EditMode safety: AddComponent fires Awake in the editor too, and
    // the headless gate must control the mount (MountOn) - self-rig is
    // the Play-path only.
    if (Application.isPlaying && _canvas == null) BuildAll();
}
/// <summary>Public entry for the headless gate (edit-mode mount onto
/// an existing rig canvas; Awake is play-only).</summary>
public void MountOn(Canvas rigCanvas)
{
    target = rigCanvas;
    if (_canvas == null) BuildAll();
}
void BuildAll()
{
    _cfg = LoadCfg();
    _square = MakeSquareSprite();
    _curtain = MakeCurtainSprite();
    if (target != null)
    {
        _canvas = target;
    }
    else
    {
        // Play-mode self rig: camera + world-space canvas + events.
        var camGo = new GameObject("UIDemoCamera");
        camGo.transform.SetParent(transform, false);
        var cam = camGo.AddComponent<Camera>();
        cam.orthographic = true;
        cam.orthographicSize = 960f;
        cam.aspect = 1080f / 1920f;
        cam.nearClipPlane = 0.1f;
        cam.farClipPlane = 500f;
        cam.transform.position = new Vector3(0f, 0f, -100f);
        cam.cullingMask = 1 <<< 5;
        cam.clearFlags = CameraClearFlags.SolidColor;
        cam.backgroundColor = new Color(0.30f, 0.30f, 0.35f);
        var cGo = new GameObject("UIDemoCanvas");
        cGo.transform.SetParent(transform, false);
        cGo.layer = 5;
        _canvas = cGo.AddComponent<Canvas>();
        _canvas.renderMode = RenderMode.WorldSpace;
        _canvas.worldCamera = cam;
        ((RectTransform)cGo.transform).sizeDelta = new Vector2(1080f, 1920f);
        if (FindObjectOfType<EventSystem>() == null)
        {
```

```
        var es = new GameObject("EventSystem", typeof(EventSystem), typeof(StandaloneInputModule));
        es.transform.SetParent(transform, false);
    }
}
for (int i = 0; i < LayerNames.Length; i++)
{
    var layer = MakeLayer(LayerNames[i]);
    _layers.Add(layer);
    layerOrder.Add(LayerNames[i]);
}
BuildBg();
BuildHud();
BuildPanel();
BuildPopup();
BuildToast();
BuildGuide();
BuildTop();
}
UIDemoConfig LoadCfg()
{
    var asset = Resources.Load<TextAsset>("cfg_uidemo");
    if (asset == null) throw new System.Exception("resource missing: cfg_uidemo");
    var cfg = JsonUtility.FromJson<UIDemoConfig>(asset.text);
    if (cfg == null) throw new System.Exception("resource parse failed: cfg_uidemo");
    return cfg;
}
RectTransform MakeLayer(string name)
{
    var go = new GameObject(name, typeof(RectTransform));
    go.transform.SetParent(_canvas.transform, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = Vector2.zero;
    rt.anchorMax = Vector2.one;
    rt.anchoredPosition = Vector2.zero;
    rt.sizeDelta = Vector2.zero;
    return rt;
}
void BuildBg()
{
    var img = _layers[0].gameObject.AddComponent<Image>();
    img.sprite = _curtain;
    img.raycastTarget = false;    // layer law: stage art never eats taps
}
void BuildHud()
{
    var strip = MakeRect(_layers[1], "HudStrip", new Vector2(0f, 1f), new Vector2(1f, 1f),
        new Vector2(0.5f, 1f), Vector2.zero, new Vector2(0f, 150f));
    var stripImg = strip.GetComponent<Image>();
    stripImg.sprite = _square;
    stripImg.color = new Color(Ink.r, Ink.g, Ink.b, 0.86f);
}
```

```

stripImg.raycastTarget = false;
var title = MakeText(_layers[1], _cfg.screenTitle, 40, new Vector2(-180f, 885f),
    TextAnchor.MiddleRight, 680f, 80f);
title.color = PaperWhite;
// Currency chip (icon + value) - real catalog art, two-state law.
var chip = MakeRect(_layers[1], "HudChip", Vector2.one * 0.5f, Vector2.one * 0.5f,
    Vector2.one * 0.5f, new Vector2(330f, 885f), new Vector2(320f, 96f));
var chipImg = chip.GetComponent<Image>();
chipImg.sprite = _square;
chipImg.color = new Color(Ink.r, Ink.g, Ink.b, 0.6f);
var chipIcon = MakeIcon(chip, _cfg.hudChipLabel == null ? null : CardIconAt(3), new Vector2(-95f, 0f), 64f);
if (chipIcon != null) chipIcon.raycastTarget = false;
var chipText = MakeText(chip.transform, _cfg.hudChipLabel, 30, new Vector2(35f, 0f),
    TextAnchor.MiddleLeft, 220f, 70f);
chipText.color = WarmYellow;
var hint = MakeText(_layers[1], _cfg.hudHint, 24, new Vector2(0f, 750f),
    TextAnchor.MiddleCenter, 900f, 50f);
hint.color = new Color(PaperWhite.r, PaperWhite.g, PaperWhite.b, 0.55f);
hint.raycastTarget = false;
}
void BuildPanel()
{
    var frameRt = MakeRect(_layers[2], "PanelFrame", Vector2.one * 0.5f, Vector2.one * 0.5f,
        Vector2.one * 0.5f, new Vector2(0f, -70f), new Vector2(1000f, 1330f));
    PanelFrame = frameRt.GetComponent<Image>();
    PanelFrame.sprite = ResolveArt(FrameStem);
    if (PanelFrame.sprite == null)
        PanelFrame.color = new Color(Ink.r, Ink.g, Ink.b, 0.35f); // X027: translucent placeholder, never an opa
    PanelFrame.type = PanelFrame.sprite == null ? Image.Type.Simple : Image.Type.Sliced;
    PanelFrame.raycastTarget = false; // X451 law: frame decor is dead
    var header = MakeText(_layers[2], _cfg.panelHeader, 52, new Vector2(0f, 500f),
        TextAnchor.MiddleCenter, 900f, 90f);
    header.color = WarmYellow;
    // 2x2 clue cards on the panel body.
    Vector2[] pos = { new Vector2(-250f, 250f), new Vector2(250f, 250f),
        new Vector2(-250f, -30f), new Vector2(250f, -30f) };
    for (int i = 0; i < pos.Length && i < _cfg.cards.Count; i++)
    {
        var card = _cfg.cards[i];
        var rt = MakeRect(_layers[2], "Card_" + i, Vector2.one * 0.5f, Vector2.one * 0.5f,
            Vector2.one * 0.5f, pos[i] + new Vector2(0f, -70f), new Vector2(460f, 270f));
        var img = rt.GetComponent<Image>();
        img.sprite = _square;
        img.color = new Color(Ink.r, Ink.g, Ink.b, 0.55f);
        var icon = MakeIcon(rt.transform, card.icon, new Vector2(-130f, 25f), 150f);
        if (icon != null) icon.raycastTarget = false;
        var label = MakeText(rt.transform, card.name, 34, new Vector2(75f, -75f),
            TextAnchor.MiddleLeft, 330f, 80f);
        label.color = PaperWhite;
        var btn = rt.gameObject.AddComponent<Button>();
    }
}

```

```

        int idx = i;
        btn.onClick.AddListener(delegate { ShowToast(_cfg.toastClue + " · " + card.name); MarkCard(idx); });
        CardButtons.Add(btn);
    }
    // Bottom action bar: primary (unique warm) + secondary (neutral).
    var prim = MakeRect(_layers[2], "PrimaryButton", Vector2.one * 0.5f, Vector2.one * 0.5f,
        Vector2.one * 0.5f, new Vector2(-260f, -520f), new Vector2(450f, 130f));
    var primImg = prim.GetComponent<Image>();
    primImg.sprite = _square;
    primImg.color = WarmYellow;
    PrimaryButton = prim.gameObject.AddComponent<Button>();
    var primLabel = MakeText(prim.transform, _cfg.primaryLabel, 44, Vector2.zero,
        TextAnchor.MiddleCenter, 430f, 110f);
    primLabel.color = Ink;
    primLabel.raycastTarget = false;
    PrimaryButton.onClick.AddListener(OpenPopup);
    var sec = MakeRect(_layers[2], "SecondaryButton", Vector2.one * 0.5f, Vector2.one * 0.5f,
        Vector2.one * 0.5f, new Vector2(260f, -520f), new Vector2(450f, 130f));
    var secImg = sec.GetComponent<Image>();
    secImg.sprite = _square;
    secImg.color = new Color(Ink.r, Ink.g, Ink.b, 0.65f);
    SecondaryButton = sec.gameObject.AddComponent<Button>();
    var secLabel = MakeText(sec.transform, _cfg.secondaryLabel, 40, Vector2.zero,
        TextAnchor.MiddleCenter, 430f, 110f);
    secLabel.color = PaperWhite;
    secLabel.raycastTarget = false;
    SecondaryButton.onClick.AddListener(delegate { ShowToast(_cfg.toastSecondary); });
}
void BuildPopup()
{
    PopupRoot = _layers[3].gameObject;
    PopupRoot.SetActive(false);
    PopupMask = _layers[3].gameObject.AddComponent<Image>();
    PopupMask.sprite = _square;
    PopupMask.color = new Color(0f, 0f, 0f, 0.55f);
    PopupMask.raycastTarget = true; // layer law: modal mask blocks
    var dlg = MakeRect(_layers[3], "PopupDialog", Vector2.one * 0.5f, Vector2.one * 0.5f,
        Vector2.one * 0.5f, new Vector2(0f, 30f), new Vector2(900f, 560f));
    PopupDialog = dlg.gameObject;
    var dlgImg = dlg.GetComponent<Image>();
    dlgImg.sprite = ResolveArt(FrameStem);
    if (dlgImg.sprite == null)
        dlgImg.color = new Color(Ink.r, Ink.g, Ink.b, 0.55f); // placeholder law, see PanelFrame
    dlgImg.type = dlgImg.sprite == null ? Image.Type.Simple : Image.Type.Sliced;
    dlgImg.raycastTarget = false;
    var title = MakeText(dlg.transform, _cfg.popupTitle, 48, new Vector2(0f, 190f),
        TextAnchor.MiddleCenter, 820f, 90f);
    title.color = WarmYellow;
    var body = MakeText(dlg.transform, _cfg.popupBody, 38, new Vector2(0f, 40f),
        TextAnchor.MiddleCenter, 780f, 160f);

```

```
body.color = PaperWhite;
var ok = MakeRect(dlg, "PopupConfirm", Vector2.one * 0.5f, Vector2.one * 0.5f,
    Vector2.one * 0.5f, new Vector2(-200f, -180f), new Vector2(340f, 120f));
var okImg = ok.GetComponent<Image>();
okImg.sprite = _square;
okImg.color = WarmYellow;
var okBtn = ok.gameObject.AddComponent<Button>();
var okLabel = MakeText(ok.transform, _cfg.popupConfirm, 40, Vector2.zero,
    TextAnchor.MiddleCenter, 320f, 100f);
okLabel.color = Ink;
okLabel.raycastTarget = false;
okBtn.onClick.AddListener(delegate { ClosePopup(); ShowToast(_cfg.toastConfirm); });
var cancel = MakeRect(dlg, "PopupCancel", Vector2.one * 0.5f, Vector2.one * 0.5f,
    Vector2.one * 0.5f, new Vector2(200f, -180f), new Vector2(340f, 120f));
var cancelImg = cancel.GetComponent<Image>();
cancelImg.sprite = _square;
cancelImg.color = new Color(Ink.r, Ink.g, Ink.b, 0.65f);
var cancelBtn = cancel.gameObject.AddComponent<Button>();
var cancelLabel = MakeText(cancel.transform, _cfg.popupCancel, 40, Vector2.zero,
    TextAnchor.MiddleCenter, 320f, 100f);
cancelLabel.color = PaperWhite;
cancelLabel.raycastTarget = false;
cancelBtn.onClick.AddListener(ClosePopup);
}
void BuildToast()
{
    var strip = MakeRect(_layers[4], "ToastStrip", Vector2.one * 0.5f, Vector2.one * 0.5f,
        Vector2.one * 0.5f, new Vector2(0f, 700f), new Vector2(900f, 120f));
    ToastStrip = strip.GetComponent<Image>();
    ToastStrip.sprite = _square;
    ToastStrip.color = new Color(Ink.r, Ink.g, Ink.b, 0.88f);
    ToastStrip.raycastTarget = false;
    ToastText = MakeText(strip.transform, "", 34, Vector2.zero,
        TextAnchor.MiddleCenter, 860f, 100f);
    ToastText.color = PaperWhite;
    ToastText.raycastTarget = false;
    strip.gameObject.SetActive(false);
}
void BuildGuide()
{
    GuideRoot = _layers[5].gameObject;
    GuideRoot.SetActive(false);
    var ring = MakeRect(_layers[5], "GuideRing", Vector2.one * 0.5f, Vector2.one * 0.5f,
        Vector2.one * 0.5f, new Vector2(-260f, -590f), new Vector2(560f, 230f));
    var ringImg = ring.GetComponent<Image>();
    ringImg.sprite = _square;
    ringImg.color = new Color(WarmYellow.r, WarmYellow.g, WarmYellow.b, 0.30f);
    ringImg.raycastTarget = false; // demo guide: non-blocking
    var label = MakeText(_layers[5], _cfg.guideText, 40, new Vector2(140f, -590f),
        TextAnchor.MiddleLeft, 500f, 90f);
```

```

        label.color = WarmYellow;
        label.raycastTarget = false;
    }
    void BuildTop()
    {
        TopChipText = MakeText(_layers[6], _cfg.topChip, 22, new Vector2(-290f, 790f),
            TextAnchor.MiddleLeft, 500f, 50f);
        TopChipText.color = new Color(PaperWhite.r, PaperWhite.g, PaperWhite.b, 0.6f);
        TopChipText.raycastTarget = false;
        // Top-layer toggle: flips the guide layer from anywhere.
        var tog = MakeRect(_layers[6], "GuideToggle", Vector2.one * 0.5f, Vector2.one * 0.5f,
            Vector2.one * 0.5f, new Vector2(455f, 790f), new Vector2(170f, 70f));
        var togImg = tog.GetComponent<Image>();
        togImg.sprite = _square;
        togImg.color = new Color(Ink.r, Ink.g, Ink.b, 0.7f);
        var togBtn = tog.gameObject.AddComponent<Button>();
        var togLabel = MakeText(tog.transform, _cfg.guideToggle, 26, Vector2.zero,
            TextAnchor.MiddleCenter, 160f, 60f);
        togLabel.color = PaperWhite;
        togLabel.raycastTarget = false;
        togBtn.onClick.AddListener(delegate { ToggleGuide(!GuideRoot.activeSelf); });
    }
    // —— interactive faces (gate taps the same paths the player taps) ——
    public void OpenPopup()
    {
        PopupRoot.SetActive(true);
        if (Application.isPlaying)
        {
            StopAllCoroutines();
            StartCoroutine(PopupScaleIn());
        }
    }
    public void ClosePopup() { PopupRoot.SetActive(false); }
    public void ShowToast(string message)
    {
        ToastText.text = message;
        ToastStrip.gameObject.SetActive(true);
        if (Application.isPlaying) StartCoroutine(ToastFade());
    }
    /// <summary>Config-sourced toast (gate path: ASCII law keeps CJK out
    /// of C# sources, so the demo copy always comes from cfg_uidemo).</summary>
    public void ShowClueToast() { ShowToast(_cfg.toastClue); }
    /// <summary>Editor gates live in a predefined editor assembly that
    /// cannot see the Sy.Core package, so art-chain diagnostics route
    /// through this runtime face. Probes both load paths because the
    /// Resources route has been observed to fail headless.</summary>
    public string ArtDiag()
    {
        var cat = ArtRuntime.Catalog;
        string res = cat == null ? "resNULL" : "resEntries=" + cat.entries.Count

```

```

        + " resGet=" + (cat.Get(FrameStem) != null);
        string ad = "n/a";
    #if UNITY_EDITOR
        var so = UnityEditor.AssetDatabase.LoadAssetAtPath<ArtCatalog>("Assets/_Game/Resources/ArtCatalog.asset");
        ad = so == null ? "adNULL" : "adEntries=" + so.entries.Count
            + " adGet=" + (so.Get(FrameStem) != null);
        var untyped = UnityEditor.AssetDatabase.LoadAssetAtPath<UnityEngine.Object>("Assets/_Game/Resources/ArtCatalog.asset");
        string untypedInfo = untyped == null ? "untypedNULL" : "untypedType=" + untyped.GetType().Name;
        var directSprite = UnityEditor.AssetDatabase.LoadAssetAtPath<Sprite>("Assets/_Game/Art/Share/P08_ui_s02_chapter_frame.png");
        string spriteInfo = directSprite == null ? "spriteNULL"
            : "spriteOK border=" + (int)directSprite.border.x + "," + (int)directSprite.border.y
            + "," + (int)directSprite.border.z + "," + (int)directSprite.border.w;
        var iconMain = UnityEditor.AssetDatabase.LoadAssetAtPath<UnityEngine.Object>("Assets/_Game/Art/Icons/P08_hot_s08_paper_money.png");
        string iconMainInfo = iconMain == null ? "iconMainNULL" : "iconMain=" + iconMain.GetType().Name
            + " asSprite=" + (iconMain as Sprite != null);
        var iconAll = UnityEditor.AssetDatabase.LoadAllAssetsAtPath(
            "Assets/_Game/Art/Icons/P08_hot_s08_paper_money.png");
        int spriteSubs = 0;
        if (iconAll != null)
            for (int i = 0; i < iconAll.Length; i++)
                if (iconAll[i] is Sprite) spriteSubs++;
        ad = ad + " | " + spriteInfo + " | " + iconMainInfo + " | iconSubs=" + spriteSubs;
    #endif

    return res + " | " + ad;
}

public void ToggleGuide(bool on) { GuideRoot.SetActive(on); }
void MarkCard(int index)
{
    if (index >= 0 && index < CardButtons.Count && CardButtons[index] != null)
        CardButtons[index].image.color = new Color(WarmYellow.r, WarmYellow.g, WarmYellow.b, 0.35f);
}

IEnumerator PopupScaleIn()
{
    var rt = (RectTransform)PopupDialog.transform;
    var targetScale = Vector3.one;
    float t = 0f;
    while (t < 0.12f)
    {
        t += Time.unscaledDeltaTime;
        rt.localScale = Vector3.Lerp(new Vector3(0.92f, 0.92f, 1f), targetScale, t / 0.12f);
        yield return null;
    }
    rt.localScale = targetScale;
}

IEnumerator ToastFade()
{

```

```
float hold = 1.6f, fade = 0.4f, t = 0f;
while (t < hold) { t += Time.unscaledDeltaTime; yield return null; }
var cg = ToastStrip.GetComponent<CanvasGroup>();
if (cg == null) cg = ToastStrip.gameObject.AddComponent<CanvasGroup>();
while (t < hold + fade)
{
    t += Time.unscaledDeltaTime;
    cg.alpha = 1f - (t - hold) / fade;
    yield return null;
}
cg.alpha = 1f;
ToastStrip.gameObject.SetActive(false);
}
string CardIconAt(int index)
{
    return _cfg != null && _cfg.cards != null && index >= 0 && index < _cfg.cards.
        ? _cfg.cards[index].icon : null;
}
// —— house-style static builders ——
static RectTransform MakeRect(RectTransform parent, string name, Vector2 aMin, Vector2 aMax,
    Vector2 pivot, Vector2 pos, Vector2 size)
{
    var go = new GameObject(name, typeof(RectTransform), typeof(Image));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = aMin;
    rt.anchorMax = aMax;
    rt.pivot = pivot;
    rt.anchoredPosition = pos;
    rt.sizeDelta = size;
    return rt;
}
static Text MakeText(Transform parent, string text, int size, Vector2 pos,
    TextAnchor anchor, float w, float h)
{
    var go = new GameObject("Text", typeof(RectTransform), typeof(Text));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = new Vector2(w, h);
    var t = go.GetComponent<Text>();
    t.font = Resources.GetBuiltinResource<Font>("LegacyRuntime.ttf");
    t.text = text;
    t.fontSize = size;
    t.alignment = anchor;
    t.color = new Color(0.91f, 0.89f, 0.85f);
    t.supportRichText = true;
    return t;
}
```

```

static Image MakeIcon(Transform parent, string stem, Vector2 pos, float size)
{
    if (string.IsNullOrEmpty(stem)) return null;
    var sprite = ResolveArt(stem);
    if (sprite == null) return null;    // X027: missing art stays procedural
    var go = new GameObject("Icon_" + stem, typeof(RectTransform), typeof(Image));
    go.transform.SetParent(parent, false);
    var rt = (RectTransform)go.transform;
    rt.anchorMin = rt.anchorMax = new Vector2(0.5f, 0.5f);
    rt.anchoredPosition = pos;
    rt.sizeDelta = new Vector2(size, size);
    var img = go.GetComponent<Image>();
    img.sprite = sprite;
    img.preserveAspect = true;
    return img;
}

/// <summary>Deterministic curtain gradient (ExploreView MakeCurtainSprite
/// profile, trimmed: red field + fold stripes + top valance).</summary>
static Sprite MakeCurtainSprite()
{
    const int W = 108, H = 192;
    var tex = new Texture2D(W, H, TextureFormat.RGBA32, false);
    tex.wrapMode = TextureWrapMode.Clamp;
    var px = new Color32[W * H];
    for (int y = 0; y < H; y++)
    {
        float v = y / (float)(H - 1);
        for (int x = 0; x < W; x++)
        {
            float u = x / (float)(W - 1);
            float fold = 0.5f + 0.5f * Mathf.Sin(u * Mathf.PI * 9f);
            float r = CurtainRed.r * (0.72f + 0.28f * fold);
            float g = CurtainRed.g * (0.72f + 0.28f * fold);
            float b = CurtainRed.b * (0.72f + 0.28f * fold);
            float a = 0.45f + 0.25f * v;
            float valance = Mathf.Clamp01((v - 0.855f) / 0.045f);
            a = Mathf.Max(a, valance * 0.92f);
            float apron = Mathf.Clamp01((0.10f - v) / 0.06f);
            a = Mathf.Max(a, apron * 0.55f);
            px[y * W + x] = new Color32(
                (byte)Mathf.RoundToInt(r * 255f),
                (byte)Mathf.RoundToInt(g * 255f),
                (byte)Mathf.RoundToInt(b * 255f),
                (byte)Mathf.RoundToInt(a * 255f));
        }
    }
    tex.SetPixels32(px);
    tex.Apply(false, false);
    return Sprite.Create(tex, new Rect(0, 0, W, H), new Vector2(0.5f, 0.5f), W);
}

```

```

        static Sprite MakeSquareSprite()
        {
            var tex = new Texture2D(8, 8, TextureFormat.RGBA32, false);
            var px = tex.GetPixels32();
            for (int i = 0; i < px.Length; i++) px[i] = new Color32(255, 255, 255, 255);
            tex.SetPixels32(px);
            tex.Apply(false, false);
            return Sprite.Create(tex, new Rect(0, 0, 8, 8), new Vector2(0.5f, 0.5f), 8f);
        }
    }
}

// UiInputChainTests.cs
using System.Collections;
using NUnit.Framework;
using Sy.P08.Adventure;
using UnityEngine;
using UnityEngine.EventSystems;
using UnityEngine.InputSystem;
using UnityEngine.InputSystem.UI;
using UnityEngine.UI;
using UnityEngine.TestTools;
namespace Sy.P08.Tests
{
    /// <summary>
    /// A0 slice-1 (U262 batch A, the 12th doc input-chain truth law): the
    /// three core P08 flows driven by SYNTHESIZED touch input through the
    /// real UGUI pointer chain - EventSystem + InputSystemUIInputModule +
    /// GraphicRaycaster + Button state - instead of onClick.Invoke() seam
    /// calls. The rig is pure programmatic UGUI in the X460 RT-gate
    /// pattern (WorldSpace canvas + orthographic camera framing the full
    /// 1080x1920 design face): batchmode Game View sizes vary, so the
    /// overlay route would crop the upper seats off-canvas on small
    /// screens - the world-space rig keeps every seat tappable at any
    /// screen size. Each test builds its own EventSystem carrying the
    /// InputSystem module, then taps real seats at their runtime screen
    /// points. Flows: chapter tap (S02 routing intent), archive card
    /// (reading overlay open/close), unlock enter (S07 gate seat
    /// liveness). Faces are asserted against the shipped config verbatim
    /// (no case text is hardcoded here).
    /// </summary>
    [TestFixture]
    public sealed class UiInputChainTests : InputTestFixture
    {
        GameObject _root;
        Canvas _canvas;
        Camera _cam;
        ChapterConfig _chapter;
        ArchiveBook _book;
        ArchiveView _archiveView;
        UnlockView _unlockView;
    }
}

```

```

public override void Setup()
{
    base.Setup();
    _root = new GameObject("UiInputChainRig");
    var esGo = new GameObject("TestEventSystem");
    esGo.transform.SetParent(_root.transform, false);
    esGo.AddComponent<EventSystem>();
    esGo.AddComponent<InputSystemUIInputModule>();
    var camGo = new GameObject("TestCamera");
    camGo.transform.SetParent(_root.transform, false);
    _cam = camGo.AddComponent<Camera>();
    _cam.orthographic = true;
    _cam.orthographicSize = 960f; // half the 1920 design height
    _cam.transform.position = new Vector3(0f, 0f, -1000f);
    _cam.farClipPlane = 2000f;
    var canvasGo = new GameObject("TestCanvas");
    canvasGo.transform.SetParent(_root.transform, false);
    _canvas = canvasGo.AddComponent<Canvas>();
    _canvas.renderMode = RenderMode.WorldSpace;
    _canvas.worldCamera = _cam;
    var canvasRt = canvasGo.GetComponent<RectTransform>();
    canvasRt.sizeDelta = new Vector2(1080f, 1920f);
    canvasGo.AddComponent<GraphicRaycaster>();
}

public override void TearDown()
{
    if (_root != null) Object.DestroyImmediate(_root);
    _root = null;
    _canvas = null;
    _cam = null;
    _chapter = null;
    _book = null;
    _archiveView = null;
    _unlockView = null;
    base.TearDown();
}

// —— flow 1: chapter tap routes through the pointer chain ——
[UnityTest]
public IEnumerator ChapterFlow_InjectedTouch_WalksRealPointerChain()
{
    BuildArchiveFace();
    string tappedId = null;
    _archiveView.ChapterTapped += id => tappedId = id;
    // The shipped live seat: a synthesized touch at the seat's
    // runtime screen point fires the navigation intent with the
    // shipped chapter id - the same walk the player's finger takes.
    Assert.IsTrue(_archiveView.IsChapterOneLive);
    yield return TouchTap(FindRect("ChapterOne"));
    Assert.AreEqual(_chapter.chapterId, tappedId);
    // The locked future chapter stays a dead face through the

```

```

        // same chain: non-interactable seats swallow the pointer.
        tappedId = null;
        Assert.IsTrue(_archiveView.IsChapterTwoLocked);
        yield return TouchTap(FindRect("ChapterTwo"));
        Assert.IsNull(tappedId);
    }
    // —— flow 2: archive card opens/closes the reading overlay ——
    [UnityTest]
    public IEnumerator ArchiveCardFlow_InjectedTouch_OpensAndClosesReadingOverlay()
    {
        BuildArchiveFace();
        // Cards ship clue-gated: unlock card 0 the way the explore
        // loop does, then tap its slot through the real chain.
        var card = _book.Cards[0];
        Assert.IsFalse(_book.IsUnlocked(card.cardId));
        Assert.IsTrue(_book.NotifyClueFound(card.unlockRef));
        yield return TouchTap(FindRect("CardSlot_0"));
        Assert.IsTrue(_archiveView.IsDetailOpen);
        Assert.AreEqual(card.title, _archiveView.DetailTitleText);
        Assert.AreEqual(card.category, _archiveView.DetailCategoryText);
        Assert.Greater(_archiveView.DetailColumnCount, 0);
        // Tap the aged-paper overlay itself to close (same chain).
        yield return TouchTap(FindRect("DetailOverlay"));
        Assert.IsFalse(_archiveView.IsDetailOpen);
    }
    // —— flow 3: unlock-enter seat fires only while it is live ——
    [UnityTest]
    public IEnumerator UnlockEnterFlow_InjectedTouch_FiresOnlyWhileSeatLive()
    {
        BuildUnlockFace(ChapterUnlockGate.StatusFreePass);
        bool entered = false;
        _unlockView.EnterTapped += () => entered = true;
        // Free-pass face: the enter seat is live and the injected
        // tap routes it.
        Assert.IsTrue(_unlockView.IsEnterLive);
        yield return TouchTap(FindRect("EnterChapter"));
        Assert.IsTrue(entered);
        // Locked face: the enter seat hides - a touch at the old
        // rect hits nothing interactive and the intent stays quiet.
        _unlockView.SyncFace(MakeFace(ChapterUnlockGate.StatusLocked));
        Assert.IsFalse(_unlockView.IsEnterLive);
        entered = false;
        yield return TouchTap(FindRect("EnterChapter"));
        Assert.IsFalse(entered);
    }
    // —— rig helpers ——
    void BuildArchiveFace()
    {
        var asset = Resources.Load<TextAsset>("cfg_chapter_case01");
        Assert.NotNull(asset, "runtime config missing: cfg_chapter_case01");
    }

```

```

        _chapter = JsonUtility.FromJson<ChapterConfig>(asset.text);
        Assert.NotNull(_chapter);
        _book = new ArchiveBook(_chapter);
        var host = new GameObject("ArchiveViewHost");
        host.transform.SetParent(_root.transform, false);
        _archiveView = host.AddComponent<ArchiveView>();
        _archiveView.Initialize(_canvas, _book);
    }
    void BuildUnlockFace(string status)
    {
        var host = new GameObject("UnlockViewHost");
        host.transform.SetParent(_root.transform, false);
        _unlockView = host.AddComponent<UnlockView>();
        _unlockView.Initialize(_canvas, "case02");
        _unlockView.SyncFace(MakeFace(status));
    }
    static ChapterUnlockGate.Face MakeFace(string status)
    {
        return new ChapterUnlockGate.Face
        {
            status = status,
            adsWatched = 0,
            adsPerUnlock = 2,
            daysRemaining = 2,
            yindeCost = 30,
        };
    }
    RectTransform FindRect(string childName)
    {
        var t = _canvas.transform.Find(childName);
        Assert.NotNull(t, "canvas child missing: " + childName);
        return (RectTransform)t;
    }
    /// <summary>One synthesized finger tap at the target's runtime
    /// screen point: touch down, one frame, touch up, then two frames
    /// so the pointer chain (module update -> EventSystem -> onClick)
    /// settles before the caller asserts.</summary>
    IEnumerator TouchTap(RectTransform target)
    {
        var screenPos = RectTransformUtility.WorldToScreenPoint(_cam, target.position);
        Assert.Greater(Screen.width, 0, "player has no screen face in this run");
        Assert.Greater(Screen.height, 0, "player has no screen face in this run");
        BeginTouch(1, screenPos);
        yield return null;
        EndTouch(1, screenPos);
        yield return null;
        yield return null;
    }
}

```